# Sobolev Regularized Score Difference Estimation in Diffusion Models

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## Abstract

Estimating the difference of two Stein's score functions is a fundamental problem in generative modeling. In particular, score differences arise naturally in transfer learning, where the score difference provides the mechanism for adapting a pre-trained model to a new target distribution, and in diffusion model-based post-training methods such as discriminator guidance. Existing estimators for score differences in these settings either lack of statistical consistency or are difficult to scale up in high-dimensions.

We propose a statistically consistent and scalable estimator for score differences based on Sobolev regularization, which plays a crucial role in ensuring consistency and stablizing the training in the small-sample regime. Mathematically, we establish a convergence rate of $\tilde{\mathcal{O}}(n^{-\frac{s-1}{d+2s-2}})$ where $d$ is the dimension and $s$ denotes the smoothness of the underlying densities, and provide a minimax lower bound of $\tilde{\Omega}(n^{-\frac{2(s-1)}{d+2s}})$ (in mean-squared error). Empirically, our estimator exhibits significantly improved stability in small-sample regimes compared to existing methods. We demonstrate its effectiveness on real-world tasks, including transfer learning for ECG signal generation, where it substantially outperforms non-regularized score difference estimators in downstream classification performance.

## 1. Introduction

The difference of Stein's score functions is defined as the gradient of the log-density ratio $\nabla \log q(\cdot) - \nabla \log p(\cdot)$ between a target distribution $q$ and a source distribution $p$. Conceptually, this difference represents the driving force required to transport samples from a source distribution $p$ to a target distribution $q$. As a result, the score difference emerges as a fundamental primitive in transfer learning for modern generative modeling (Liu et al., 2023; Ouyang et al., 2024; Wang et al., 2024).

This estimation problem is central in post-training of diffusion models, which adapts pre-trained generative models to align with human preferences, structural constraints, or downstream tasks. A wide range of approaches have been proposed, including RLHF (Black et al., 2023; Fan et al., 2023), stochastic control–based formulations (Tang & Zhou, 2024; Han et al., 2024b; Uehara et al., 2024), and classifier-guided or conditioning-based methods. Notably, most of these can be unified as add-on mechanisms to pre-trained dynamics:

$$\mathrm{d}Y_t = s_\theta(t, Y_t)\mathrm{d}t - h_\eta(t, Y_t)\mathrm{d}t + \sigma(t)\mathrm{d}W_t, \quad (1)$$

where $s_\theta(t,\cdot)$ is the pre-trained score function approximating $\nabla \log p_t(\cdot)$, and $h_\eta(t,\cdot)$ is an additive control term. In frameworks such as discriminator guidance and conditional generation, via Doob's $h$-transform, the term $h_\eta(t,\cdot)$ approximates $\nabla \log q_t(\cdot)$, where $q_t$ is a target distribution carrying information from constraints, classifiers, or preference signals (Denker et al., 2024; Du et al., 2024; Pidstrigach et al., 2025; Howard et al., 2025). In diffusion models, post-training is a form of transfer learning, transferring pre-training knowledge to new data-generation tasks. For simplicity, we use "transfer learning" to refer to this broader setting throughout the paper.

In transfer learning, the target task typically has far fewer samples than the source task, rendering direct estimation of the target score $\nabla q_t$ unstable. Fortunately, because the target and source tasks are closely related, the density ratio $\frac{q_t}{p_t}$ often exhibits exploitable structure. This motivates directly estimating the density ratio and its gradient, since

$$\nabla \log q_t(\cdot) - \nabla \log p_t(\cdot) = \frac{\nabla (q_t/p_t)(\cdot)}{q_t/p_t(\cdot)}. \quad (2)$$

Unlike estimating and differencing the two scores separately, this approach exploits shared geometry and is more structure-aware and data-efficient in low-sample regimes (see Figure 1).

Despite arising naturally in many settings, obtaining a reliable estimator for this score difference remains challenging. The most prevalent large-scale approaches (Ouyang et al.,

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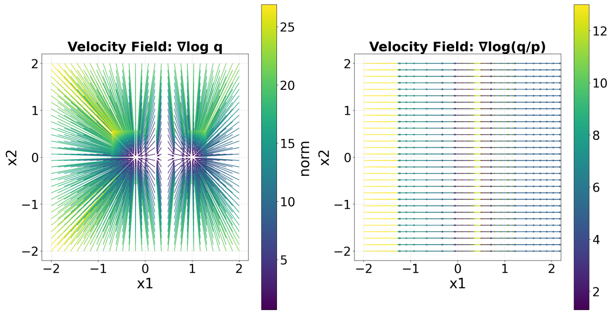


*Figure 1.* **Visualization of the score functions**. The score difference field (right) is markedly smoother and less variable than the absolute target score (left), demonstrating the structural benefit of directly estimating the score difference.

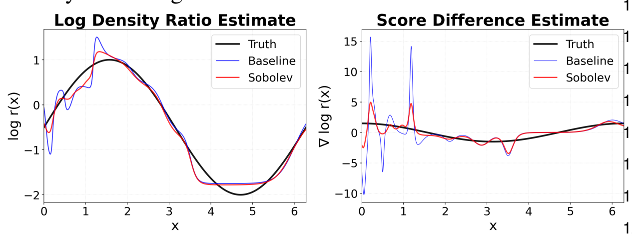


*Figure 2.* **Impact of high-frequency noise.** Comparison of $\log r$ (left) and $\nabla \log r$ (right) estimates. Standard classification (blue) captures the function value but fails on the gradient due to overfitting. Our Sobolev-regularized method (red) enforces smoothness, recovering the gradient accurately by filtering out noise.

2024; Kim et al., 2022) adopt a "classify-then-differentiate" pipeline: first estimating the log-density ratio via binary classification, and then differentiating the fitted model. However, this approach lacks theoretical guarantees on gradient convergence. As illustrated in Figure 2, standard classifiers tend to overfit high-frequency but low-amplitude noise components in the data. This overfitting yields small ratio errors (left) but catastrophic oscillations after differentiation (right).

We address this challenge via Sobolev regularization, which controls the smoothness of the log-density ratio gradient and prevents overfitting to spurious high-frequency noise. With $n$ samples from both the source $p$ and target $q$, we prove a convergence rate of $\tilde{\mathcal{O}}(n^{-\frac{s-1}{d+2s-2}})$ for the score difference error (Theorem 3.1, 5.2), where $d$ is the dimension and $s$ is the smoothness parameter. We also provide a minimax lower bound of $\tilde{\Omega}(n^{-\frac{2(s-1)}{d+2s}})$ (Theorem 4.1), showing near-optimality. Experiments verify these findings across diverse tasks, including joint domain adaptation via Wasserstein gradient flow (Sec 6.2.1) and transfer learning for diffusion models (Sec 6.2.2), confirming that Sobolev regularization yields stable and consistent score difference estimators.

**Closely related literature.** Our work is closely related to two lines of literature: score difference estimation and Sobolev regularization in machine learning.

Score difference estimation underlies a broad class of applications related to diffusion models. For example, (Ouyang et al., 2024) propose using score difference estimation for transfer learning tasks in diffusion models, which is later refined by (Wang et al., 2024) with an additional residual fine-tuning step. For post-training problems, the discriminator guidance framework of (Kim et al., 2022) highlights the central role of score difference estimation. In this approach, a discriminator is trained to distinguish samples from the target distribution and the pre-trained model, and its gradient provides an estimate of the score difference $\nabla \log q_t - \nabla \log p_t$, which is then injected as a control term into the diffusion dynamics. This framework was further extended to noisy data settings in (Cong et al., 2025). However, the above papers primarily emphasize methodological development and empirical validation, while theoretical convergence guarantees remain unestablished. On the other hand, several alternative approaches provide convergence guarantees but may still suffer from practical limitations such as scalability and numerical stability. For example, assuming that the ground-truth score difference lies in a reproducing kernel Hilbert space (RKHS), (Srikanth et al.) derive a closed-form kernel-based solution. However, this approach incurs prohibitive computational costs, as it requires evaluating kernel distances against the entire training dataset at each inference step, rendering it impractical at scale. Similarly, (Liu et al., 2023) propose a consistent score difference estimator for Wasserstein gradient flow (WGF) methods, but their local kernel approach faces the same scalability challenges. More recently, (Verine et al., 2025) address objective bias in discriminator guidance by adopting a score-matching objective in which the regression target is the residual error of a pre-trained score. However, without explicit regularity constraints, the resulting estimator can be dominated by variance, particularly in low-density regions or when the pre-trained score is already relatively accurate. In contrast to these approaches, our Sobolev-regularized score difference estimator retains the computational efficiency of classification-based frameworks, while offering enhanced consistency and robustness via the introduction of a Sobolev regularization term.

Highlighting the central role of regularization, (Husain & Nock) point out that discriminators in guidance frameworks must be well regularized to ensure generalization, a perspective that closely aligns with our work. Sobolev-type regularization, often realized through gradient penalties, has been widely studied in the literature on generative adversarial networks (GANs) (Lin et al., 2025; Roth et al., 2017; Mescheder et al., 2018), whereas its role in diffusion models remains comparatively underexplored. In these settings, the discriminator can be interpreted as estimating a time-varying density ratio. Moreover, the underlying motivations differ substantially. In those works, gradient penalties are introduced to promote dynamical stability by shifting the eigenvalues of the Jacobian and preventing oscillations around local Nash equilibria. In contrast, our motivation is rooted in statistical estimation: we employ Sobolev regularization to constrain the hypothesis space, ensuring that the estimated score difference remains accurate and robust to overfitting to high-frequency noise. Finally, beyond generative modeling, (Ding et al., 2025) study Sobolev-penalized deep semi-supervised regression for the joint estimation of a regression function and its gradient, leveraging a large set of unlabeled covariates to approximate the Sobolev penalty.

## 2. Problem setup and proposed method

Our goal is to formalize transfer learning from a source $p$ to a target $q$, by estimating the score difference in a way that is structure-aware, data-efficient, and stable.

To focus on the essential building block of our framework, we first consider a bounded state space. The extension to unbounded domains and the inclusion of a time component, tailored to diffusion models, are deferred to Section 5, where the generalization is straightforward.

Let $[0,1]^d \subseteq \Omega \subseteq \mathbb{R}^d$, $d \in \mathbb{N}$ be a bounded and connected domain with sufficiently smooth boundary $\partial\Omega$. Let $P, Q \in \mathcal{P}(\Omega)$ be absolute continuous with respect to Lebesgue measure with densities $p, q$. For any $x \in \Omega$ let

$$\rho(x) := \tfrac{1}{2}p(x) + \tfrac{1}{2}q(x), \qquad \mu(\mathrm{d}x) := \rho(x)\,\mathrm{d}x,$$

and define the log-density ratio

$$f^\star(x) := \log \frac{q(x)}{p(x)}. \tag{3}$$

For a multi–index $\alpha = (\alpha_0, \alpha_1, \dots, \alpha_d) \in \mathbb{N}^{1+d}$, denote $|\alpha| := \alpha_0 + \cdots + \alpha_d$ and $D^\alpha f := \frac{\partial^{|\alpha|} f}{\partial t^{\alpha_0} \partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}$, interpreted in the weak (distributional) sense. For an integer $s \geq 0$, the weighted Sobolev space $H^s(\mu)$ is given by

$$H^s(\mu) := \left\{ f \in L^2(\mu) \;\middle|\; \begin{array}{l} D^\alpha f \in L^2(\mu), \text{for all} \\ \quad \text{multi-indices } \alpha \text{ with } |\alpha| \leq s \end{array} \right\}.$$

For the supremum norm, we denote the $L^\infty(\mu)$ norm as the essential supremum with respect to the measure $\mu$:

$$\|f\|_{L^\infty(\mu)} := \inf\big\{C \geq 0 : \mu(\{x \in \Omega : |f(x)| > C\}) = 0\big\}.$$

Let $\mu_0$ be the uniform distribution on $\Omega$. We assume the following outstanding assumptions.

**Assumption 2.1** (Density functions)**.** Assume $p$ and $q$ satisfy the following conditions:

1. $p, q \in C^2(\Omega)$ and are strictly positive on $\Omega$.
2. The following Neumann boundary condition holds: $\rho(x)\,\partial_n f^\star(x) = 0 \quad \text{for all } x \in \partial\Omega$.
3. Fix $s \geq 2$. We assume $f^\star \in H^s(\mu_0)$.

Given that $p, q$ are positive and $C^2$ on $\Omega$, we have $f^\star \in C^2(\Omega)$. Hence, there exists $M^\star > 0$ such that

$$\max\{\|f^\star\|_{L^\infty(\mu_0)}, \|\nabla f^\star\|_{L^\infty(\mu_0)}\} \leq M^\star.$$

Motivated by the boundedness of the true density ratio $f^\star$, we restrict the function class to ensure stability in the optimization. Specially, fix a constant $M \geq M^\star$, we define a bounded Sobolev subset of $\mathcal{H}^s(\mu_0)$ as follows:

$$\mathcal{H}_M := \left\{ f \in H^1(\mu_0) : \|f\|_{L^\infty(\mu_0)} \leq M \right\}. \tag{4}$$

By construction, this set is closed and convex in $H^1(\mu_0)$, and it contains the ground truth $f^\star$.

**Learning objective.** Our objective is to estimate the score difference

$$\nabla_x f^\star(x) := \nabla_x \log \frac{q(x)}{p(x)},$$

given access only to finite samples:

$$\mathcal{S} = \{X_i\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} p, \;\; \mathcal{T} = \{X_i\}_{i=n+1}^{2n} \overset{\text{i.i.d.}}{\sim} q. \tag{5}$$

When $p$ and $q$ share similar geometric structure, the vector field $\nabla \log \frac{q(x)}{p(x)}$ often concentrates on a low-dimensional manifold, making it easier to learn than the full score of $q$. The standard approach first estimates the log-density ratio and then differentiates it directly (Ouyang et al., 2024; Kim et al., 2022). Usually, the estimation of log-density ratio is trained via classification loss.

**Binary classification reformulation.** We introduce a label-augmented mixture model, offering a convenient probabilistic representation for learning the log-density ratio. Specifically, denote by $\mu_{p,q}$ the joint law of $(X, Y)$ with

$$\begin{aligned} &Y \sim \text{Bernoulli}(1/2), \\ &X \mid (Y=1) \sim q, \; X \mid (Y=0) \sim p. \end{aligned} \tag{6}$$

Consider the dataset in (5). Let $Y_i = 0$ for $i = 1, \dots, n$ and $Y_i = 1$ for $i = n+1, \dots, 2n$, indicating whether $X_i$ is drawn from $\mathcal{S}$ or $\mathcal{T}$. Then $\mathcal{D} = (X_i, Y_i)_{i=1}^{2n}$ consists of i.i.d. samples from $\mu_{p,q}$.

Under such setting, the posterior classification probability is, for $x \in \Omega$,

$$\eta(x) := \mathbb{P}(Y = 1 \mid X = x) = \frac{q(x)}{p(x) + q(x)}$$

The corresponding Bayes logit is $\log \frac{\eta(x)}{1-\eta(x)} = \log \frac{q(x)}{p(x)} = f^\star(x)$. Therefore, if we let $\sigma(u) := \frac{1}{1+e^{-u}}$ and define the cross-entropy loss

$$\ell_{\mathrm{CE}}(y, u) := -y \log \sigma(u) - (1-y)\log(1-\sigma(u)),$$

then $f^*$ defined in (3) is the minimizer of the population cross-entropy risk

$$L_{\mathrm{CE}}(f) := \mathbb{E}_{(X,Y)\sim\mu_{p,q}}\left[\,\ell_{\mathrm{CE}}(Y, f(X))\,\right]$$

among all measurable functions $f : \Omega \to \mathbb{R}$. Define the empirical cross-entropy risk as

$$\widehat{L}_{\mathrm{CE},\mathcal{D}}(f) := \frac{1}{2n}\sum_{i=1}^{2n}[\ell_{\mathrm{CE}}(Y_i, f(X_i))].$$

**Functional class: clipped and gradient-bounded sparse neural network functional space.** We consider a function class represented by sparse neural networks (Schmidt-Hieber, 2020; Suzuki, 2018; Farrell et al., 2021) and use the $\mathrm{ReLU}^3$ activation

$$\eta_3(x) := \max\{x^3, 0\},$$

motivated by its ability to represent spline-type approximants and the associated approximation theory (Lu et al., 2021). For vector inputs, $\eta_3$ is applied element-wise. Fix a constant $M > M^\star$. Let $T_M : \mathbb{R} \to [-M, M]$ be a fixed nondecreasing $C^1$ clipping map satisfying

$$|T_M(u)| \le M, \qquad |T_M'(u)| \le 1,$$

and, for some $M_0$ with $M^\star < M_0 < M$,

$$T_M(u) = u, \qquad |u| \le M_0.$$

Given the number of layers $L$, width $W$, sparsity constraint $S$, and norm constraint $B$, define the clipped and gradient-bounded neural network class as

$$\begin{aligned}
&\mathcal{F}_M(L, W, S, B) := \\
&\Big\{g(x) = T_M\Big[\big(\mathcal{W}^{(L)}\eta_3(\cdot) + b^{(L)}\big) \circ \cdots \circ \big(\mathcal{W}^{(1)}x + b^{(1)}\big)\Big] : \\
&\quad \mathcal{W}^{(1)} \in \mathbb{R}^{W\times d}, \quad \mathcal{W}^{(2,\dots,L-1)} \in \mathbb{R}^{W\times W}, \quad \mathcal{W}^{(L)} \in \mathbb{R}^{1\times W} \\
&\quad \sum_{l=1}^{L}\Big(\|\mathcal{W}^{(l)}\|_0 + \|b^{(l)}\|_0\Big) \le S, \qquad \max_l \|\mathcal{W}^{(l)}\|_\infty \vee \|b^{(l)}\|_\infty \le B, \\
&\quad \|g\|_{L^\infty(\mu)} \le M, \qquad \|\nabla g\|_{L^\infty(\mu)} \le M\Big\}.
\end{aligned} \tag{7}$$

Here $\circ$ denotes function composition, $\|\cdot\|_0$ is the $\ell_0$-norm, and $\|\cdot\|_\infty$ is the entrywise $\ell_\infty$-norm.

**Proposed learning method.** To control both function values and gradients, we define the Sobolev seminorm

$$\mathcal{R}(f) := \|\nabla f\|^2_{L^2(\mu)},$$

for any $s \ge 1$ and $f \in H^s(\mu)$. For $\lambda > 0$, the population Sobolev-regularized risk is defined as

$$J_\lambda(f) := L_{\mathrm{CE}}(f) + \lambda\,\mathcal{R}(f).$$

The existence of the unique minimizer $f_\lambda$ is provided in Lemma A.1. Define the empirical Sobolev penalizer:

$$\widehat{\mathcal{R}}_{\mathcal{D}}(f) := \frac{1}{2n}\sum_{j=1}^{2n}\|\nabla f(X_j)\|_2^2 = \frac{1}{2n}\sum_{j=1}^{2n}\sum_{k=1}^{d}\big(\partial_k f(X_j)\big)^2.$$

and the empirical energy functional:

$$\widehat{J}_{\lambda,\mathcal{D}}(f) := \widehat{L}_{\mathrm{CE},\mathcal{D}}(f) + \lambda\,\widehat{\mathcal{R}}_{\mathcal{D}}(f).$$

Our method proceeds by first estimating the log–density ratio as the minimizer of an empirical energy functional,

$$\widehat{f}_{\lambda,\mathcal{F}} \in \arg\min_{f\in\mathcal{F}} \widehat{J}_{\lambda,\mathcal{D}}(f), \tag{8}$$

and then taking its gradient to obtain the desired score difference. We assume that the minimum is attained in $\mathcal{F}$.

## 3. Upper bound analysis

In this section, we provide a statistical (estimation) convergence rate for our Soblev regularized estimator, in terms of the number of samples used in optimization, for which the proof is deferred to Appendix A.4.

**Theorem 3.1** (Convergence rate for Sobolev-regularized logistic empirical risk minimization(ERM))**.** *Suppose Assumption 2.1 hold. Consider the sparse Deep Neural Network function space $\mathcal{F}_M(L, W, S, B)$ with parameters $L = \mathcal{O}(1), W = \mathcal{O}\left(n^{\frac{d}{d+2s-2}}\right), S = \mathcal{O}\left(n^{\frac{d}{d+2s-2}}\right), B = \mathcal{O}\left(n^{\frac{d}{d+2s-2}}\right)$. Let $\lambda = \mathcal{O}(n^{-\frac{s-1}{d+2s-2}})$, then the soblev-penalized estimator $\widehat{f}_{\lambda,\mathcal{F}}$ in (8) satisfies the following upper bound with probability $1 - 2n^{-2}$:*

$$\|\widehat{f}_{\lambda,\mathcal{F}} - f^\star\|^2_{H^1(\mu)} \lesssim n^{-\frac{s-1}{d+2s-2}}\log n.$$

*Remark* 3.2. Since $\mu$ admits a $C^2$ density $\rho$ on $\Omega$, it is equivalent the uniform distribution $\mu_0$. Consequently, the above bound can also be equivalently expressed as

$$\|\widehat{f}_{\lambda,\mathcal{F}} - f^\star\|^2_{H^1(\mu_0)} \lesssim n^{-\frac{s-1}{d+2s-2}}\log n.$$

Moreover, since $J_\lambda$ is strongly convex over $\mathcal{H}_M$, a local Rademacher complexity argument yields a sharper rate than the $\mathcal{O}(n^{-\frac{s}{d+4s}})$ rate obtained for the Sobolev-regularized estimators in (Ding et al., 2025), where they used global Rademacher complexity for regression problems.

**Proof outline.** The proof proceeds in four steps. First, we establish key analytic properties of the population energy functional $J_\lambda$, including strong convexity and smoothness (see Lemma 3.3). Second, we quantify the regularization-induced bias at the population level (see Lemma 3.4). Third, we relate the generalized error to the excess energy via localized Rademacher complexity (see Theorem 3.5). Finally, combining these ingredients yields the desired rate.

For any functional $J : H^s(\mu) \to \mathbb{R}$, let $DJ(g)$ be the Fréchet derivative of $J$ at $g$. Then we have the following result.

**Lemma 3.3** (Strong convexity and smoothness of the penalized risk)**.** *Define* $c_{\min} := \frac{1}{4\cosh^2(M/2)}$*. Then the penalized risk* $J_\lambda(f) := L_{\mathrm{CE}}(f) + \lambda \mathcal{R}(f)$ *is Fréchet differentiable on* $\mathcal{H}_M$ *and satisfies, for all* $f, g \in \mathcal{H}_M$*,*

$$\begin{aligned} & J_\lambda(f) - J_\lambda(g) - DJ_\lambda(g)[f-g] \\ \geq\ & \frac{c_{\min}}{2}\|f-g\|^2_{L^2(\mu)} + \lambda\|\nabla(f-g)\|^2_{L^2(\mu)}, \quad (9) \\ \textit{and}\quad & J_\lambda(f) - J_\lambda(g) - DJ_\lambda(g)[f-g] \\ \leq\ & \|f-g\|^2_{L^2(\mu)} + \lambda\|\nabla(f-g)\|^2_{L^2(\mu)}. \quad (10) \end{aligned}$$

Note that the parameter $M > 0$ used in $c_{\min}$ is the uniform logit bound, i.e. $|f(x)| \leq M$ for all $f \in \mathcal{F} \cup f^\star$ and $\mu$-a.e. $x$ (see the details in Proposition A.2).

Our argument follows the general outline of (Ding et al., 2025), with an important modification: because the cross-entropy loss fails to be globally strongly convex, uniqueness cannot be obtained directly. Instead, we establish conditional strong convexity restricted to $\mathcal{H}_M$. Our proof proceeds by (i) deriving strictly positive pointwise curvature bounds for the logistic loss within $[-M, M]$, (ii) lifting the curvature bounds to the functional space via Taylor expansions, and (iii) combining the bounds with the quadratic structure of the Sobolev regularizer.

Denote $\Delta$ as the Laplacian operator $\Delta f(x) := \sum_{j=1}^d \partial_{x_j x_j} f(x)$ , interpreted in the weak sense for $f \in H^1(\mu)$. Define

$$R_0^\star := \Delta f^\star + \nabla f^\star \cdot \nabla \log \mu \in L^2(\mu),$$

and $\beta := 64\, \|R_0^\star\|^2_{L^2(\mu)} \cosh^4\left(\frac{M}{2}\right)$. Next we quantify the regularization-induced bias.

**Lemma 3.4** (Bias of the population Sobolev solution)**.** *For all* $\lambda > 0$*, the unique minimizer* $f_\lambda$ *of* $J_\lambda$ *satisfies*

$$\|f_\lambda - f^\star\|^2_{L^2(\mu)} \leq \beta\,\lambda^2, \;\; \|\nabla(f_\lambda - f^\star)\|^2_{L^2(\mu)} \leq \beta\,\lambda.$$

The proof relies on testing the Euler-Lagrange (EL) equation with error $\delta_\lambda := f_\lambda - f^\star$. A crucial insight is to use Green's identity to relate the regularization part in EL equation to the Laplacian of $f^\star$, which controls the bias magnitude.

**Population envelope condition.** In the generalization analysis below, we use the population minimizer $f_\lambda$ as the reference function. We assume that the fixed envelope $M$ is chosen large enough so that, for the range of $\lambda$ considered,

$$\|f_\lambda\|_{L^\infty(\mu)} \leq M, \qquad \|\nabla f_\lambda\|_{L^\infty(\mu)} \leq M.$$

Equivalently, this condition may be replaced by a separate regularity lemma showing that the population Sobolev solution $f_\lambda$ satisfies the above bounds uniformly for the chosen regularization schedule.

The proof is deferred to Appendix A.3. The key property facilitating our fast rate analysis is the structural constraint of the hypothesis space $\mathcal{H}_M(\mu)$. Since optimization is performed over a bounded, convex domain, the regularized loss $J_\lambda$ is strongly convex and the estimator remains bounded.

Strong convexity and boundedness ensure that the variance of the error diminishes as the estimator approaches the optimum. This motivates the use of localized, rather than global, complexity measures. In the proof, we formalize this intuition using the framework of (Lu et al., 2021).

**Theorem 3.5** (Shifted oracle inequality for the clipped sieve)**.** *Fix* $0 < \lambda < 1$*. Let*

$$\mathcal{F} := \mathcal{F}_M(L, W, S, B)$$

*be the clipped and gradient-bounded neural-network sieve, where* $L = O(1)$*,* $W = O(N)$*,* $S = O(N)$*, and* $B = O(N)$*. Let* $f_0 \in \mathcal{F}$ *be any fixed comparator independent of the data, and* $f_\lambda$ *as the population minimizer of* $J_\lambda$*. Let*

$$\widehat{f}_{\lambda,\mathcal{F}} \in \arg\min_{f\in\mathcal{F}} \widehat{J}_{\lambda,\mathcal{D}}(f).$$

*Then for any* $t > 0$*, with probability at least* $1 - e^{-t}$*,*

$$J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - J_\lambda(f_\lambda) \lesssim J_\lambda(f_0) - J_\lambda(f_\lambda) + r^\star + \frac{t}{n}, \quad (11)$$

*where* $r^\star$ *is the critical radius of the sub-root function*

$$\phi(r) := C_M\sqrt{\frac{S3^L r}{n}\log(BWn)}.$$

*Here* $C_M$ *depends only on the fixed envelope* $M$ *and* $c_{\min}^{-1}$*, but not on* $n, N, W, S, B$*.*

## 4. Minimax lower bound

In this section, we provide a minimax lower bound for the ratio estimation, for which the proof is deferred to Appendix B.

Consider the dataset $\mathcal{D} := (X_i, Y_i)_{i=1}^{2n}$ constructed as below (6), and assume the underlying $(p, q)$ satisfies Assumption 2.1. Let $\mathcal{P}$ denote the class of all such $(p, q)$ pairs. We write $\mathbb{E}_{p,q}$ for expectation with respect to $\mathcal{D}$ under $(p, q)$.

**Theorem 4.1** (Minimax lower bound for score difference estimation)**.** *For any estimator*

$$\psi : \left(\mathbb{R}^{d+1}\right)^{2n} \to H^1(\mu_0),$$

*we have the minimax lower bound*

$$\begin{aligned} & \inf_\psi \sup_{(p,q)\in\mathcal{P}} \mathbb{E}_{p,q}\left[\left\|\psi(\{X_i, Y_i\}_{i=1}^{2n}) - f_{p,q}\right\|^2_{H^1(\mu_0)}\right] \\ & \gtrsim\ n^{-2(s-1)/(2s+d)}. \end{aligned} \quad (12)$$

**Technical novelties.** Our lower-bound analysis builds on the Local Fano method, but requires several new ingredients to address three structural obstacles specific to our setting. First, we confront a regularity mismatch between the observation and the target quantity: unlike (Lu et al., 2021), where the observation involves a differential operator that smooths the estimation problem, here we must recover high-order $H^1$ information from lower-order $L^2$ observations. Second, we characterize the information-theoretic limits under a heteroscedastic observation model induced by the joint sampling of the source and target densities. Finally, we handle the global normalization constraint intrinsic to density estimation, showing that local perturbations remain statistically indistinguishable even after enforcing normalization through global partition functions.

**Discussion on the optimality gap.** Comparing Theorem 3.1 with Theorem 4.1, we observe a gap between the achievable upper rate $\tilde{\mathcal{O}}(n^{-\frac{s-1}{d+2s-2}})$ and the minimax lower bound $\tilde{\Omega}(n^{-\frac{2(s-1)}{d+2s}})$. We conjecture that this suboptimality is an artifact of the saturation phenomenon associated with single-step Tikhonov-type regularization (Bauer et al., 2007; Engl et al., 1996). Within our framework, the Sobolev penalty $\lambda\|\nabla f\|^2$ induces a bias–variance trade-off in which the regularization parameter $\lambda$ simultaneously governs approximation error and statistical stability. In particular, enforcing conditional strong convexity requires $\lambda$ to be sufficiently large to control the stochastic error–manifested through the $1/\lambda$ factor in the gradient stability bounds. This constraint limits how rapidly the bias can decay, thereby preventing the estimator from attaining the optimal nonparametric rate.

Theoretically, the gap between the upper and lower bounds could be closed using iterative regularization schemes, such as iterated Tikhonov regularization (Engl et al., 1996), which are known to attain optimal convergence rates. However, implementing such methods in a deep learning setting would require training a sequence of neural networks where each subsequent network relies on the previous one, leading to prohibitive computational and memory costs for large-scale applications such as diffusion model and transfer learning. Consequently, while our single-step estimator is not minimax optimal, it is computationally efficient and practically scalable, and already exhibits strong empirical performance and stability in the small-sample regime (see Table 2).

# 5. Generalization to diffusion models: time dependence and unbounded domains

In this section, we extend our framework to transfer learning for diffusion models. Compared to the static setting in Section 2, two additional challenges arise: (i) the density ratio becomes time dependent; and (ii) Gaussian perturbations render the data support unbounded. We address both issues by reformulating the problem on an augmented time-space domain, applying a truncation argument and proposing a projected-rescaled algorithm.

**Augmented time-space formulation under the VP forward model.** To formalize the diffusion-model setting, we focus on the variance-preserving (VP) forward noising model. Let $X_0 \sim p_0$ and $Y_0 \sim q_0$ denote the source and target initial variables. For $t \in [0,T]$, assume

$$X_t = \alpha_t X_0 + \sigma_t \xi, \qquad Y_t = \alpha_t Y_0 + \sigma_t \xi', \qquad \xi, \xi' \sim \mathcal{N}(0, I_d),$$

where $\alpha_t \in \mathbb{R}$ and $\sigma_t > 0$ are deterministic scalar VP coefficients.

Fix $t_0 \in (0,T)$ as the early-stopping time commonly used in the diffusion model literature (Han et al., 2024a). In this context, the goal is to estimate the score difference $\nabla_x f^\star(t,x) := \nabla_x \log(q_t(x)/p_t(x))$ for $t \in [t_0, T]$. We consider the corresponding learning problem on the augmented space:

$$\tilde{\Omega} := [t_0, T] \times \mathbb{R}^d, \quad \tilde{z} := (t,x) \in \tilde{\Omega}.$$

We define the time-dependent mixture measure as $\mu_t := \frac{1}{2}(p_t + q_t)$, which serves as our reference measure on each time slice.

The objective is to learn a function $f : \tilde{\Omega} \to \mathbb{R}$ that minimizes a time-averaged risk. Since the temporal domain $[t_0, T]$ is compact, the main difficulty arises from the unbounded spatial domain $\mathbb{R}^d$. In particular, strong convexity fails globally on $\mathbb{R}^d$. To address this issue, we impose suitable assumptions on the tail behavior of the data distribution, which is common and reasonable assumptions to assume in the diffusion model literature (Han et al., 2024a; Li et al., 2023; Kong et al., 2024).

**Assumption 5.1** (Regular VP forward model with compact initial support)**.** The initial source and target distributions are compactly supported: there exists $R_{\mathrm{init}} < \infty$ such that

$$\mathrm{supp}(p_0) \cup \mathrm{supp}(q_0) \subseteq B_{R_{\mathrm{init}}}.$$

Moreover, the VP coefficients satisfy

$$\alpha, \sigma \in C^s([t_0, T]),$$

and there exist constants

$$0 < \underline{\sigma} \le \overline{\sigma} < \infty, \qquad C_{\mathrm{VP}} < \infty,$$

such that for all $t \in [t_0, T]$,

$$\underline{\sigma} \le \sigma_t \le \overline{\sigma}, \qquad \max_{0 \le a \le s} \left( |\partial_t^a \alpha_t| + |\partial_t^a \sigma_t| \right) \le C_{\mathrm{VP}}.$$

**Truncation-dependent clipped network class.** For $R > 0$, define

$$C_{2R} := [t_0, T] \times B_{2R}, \qquad C_1 := [t_0, T] \times B_1.$$

The neural network is trained on the fixed input domain $C_1$, with input dimension $d+1$. Following the clipped-gradient construction in Section 3, we use an $R$-dependent time-space class

$$\mathcal{F}_R := \mathcal{F}^{(d+1)}_{M_R}(L, W, S, B), \qquad M_R := C_M(1+R),$$

where $\mathcal{F}^{(d+1)}_{M_R}(L, W, S, B)$ denotes the clipped sparse $\mathrm{ReLU}^3$ network class with input dimension $D = d+1$ satisfying

$$\|f\|_{L^\infty(C_1)} \le M_R, \qquad \|\nabla_{\bar{x}} f\|_{L^\infty(C_1)} \le M_R.$$

Here $\bar{x} \in B_1$ denotes the rescaled spatial input. The constant $C_M$ is chosen large enough so that the rescaled target

$$f^\star_R(t, \bar{x}) := f^\star(t, 2R\bar{x})$$

and its rescaled spatial gradient $\nabla_{\bar{x}} f^\star_R$ are both bounded by $M_R$ on $C_1$. The proof of feasibility is deferred to Appendix C.2.

**Projected-rescaled empirical objective.** For $R > 0$, define the bounded time-space cylinder

$$C_{2R} := [t_0, T] \times B_{2R}, \qquad B_{2R} := \{x \in \mathbb{R}^d : \|x\| \le 2R\}.$$

Define the Euclidean projection onto $B_{2R}$ by

$$\Pi_{2R}(x) := \frac{x}{\max\{1, \|x\|/(2R)\}}, \qquad \bar{x} := \frac{\Pi_{2R}(x)}{2R} \in B_1.$$

Given $\mathcal{D}_{\mathrm{ext}} = \{(t_i, X_i, Y_i)\}_{i=1}^n$, where $t_i \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}([t_0, T])$, $Y_i \sim \mathrm{Bernoulli}(1/2)$, and

$$X_i \mid (t_i, Y_i = 0) \sim p_{t_i}, \qquad X_i \mid (t_i, Y_i = 1) \sim q_{t_i},$$

we estimate the rescaled log-density ratio by

$$\widehat{f}_R \in \arg\min_{f \in \mathcal{F}_R} \widehat{\mathcal{J}}^{\mathrm{proj}}_{\lambda, \mathcal{D}_{\mathrm{ext}}, 2R}(f),$$

where

$$\widehat{\mathcal{J}}^{\mathrm{proj}}_{\lambda, \mathcal{D}_{\mathrm{ext}}, 2R}(f) := \frac{1}{n}\sum_{i=1}^n \left[\ell_{\mathrm{CE}}\big(Y_i, f(t_i, \bar{X}_i)\big) + \frac{\lambda}{4R^2}\|\nabla_{\bar{x}} f(t_i, \bar{X}_i)\|_2^2\right], \qquad f \in \mathcal{F}_R. \tag{13}$$

The factor $1/(4R^2)$ accounts for the spatial rescaling $x = 2R\bar{x}$: if $h(t,x) = f(t, x/(2R))$, then

$$\nabla_x h(t, x) = \frac{1}{2R}\nabla_{\bar{x}} f\Big(t, \frac{x}{2R}\Big).$$

**Cutoff extension and core-tail decomposition.** Let $\chi_R : \mathbb{R}^d \to [0,1]$ be a smooth cutoff satisfying

$$\chi_R \equiv 1 \text{ on } B_R, \qquad \chi_R \equiv 0 \text{ on } \mathbb{R}^d \backslash B_{2R}, \qquad \|\nabla \chi_R\|_\infty \lesssim R^{-1}.$$

Using the optimizer $\widehat{f}_R$ from (13), define the global estimator on $\widetilde{\Omega} = [t_0, T] \times \mathbb{R}^d$ by

$$\widetilde{f}^{(R)}(t, x) := \chi_R(x)\, \widehat{f}_R\Big(t, \frac{\Pi_{2R}(x)}{2R}\Big). \tag{14}$$

On $B_R$, we have $\Pi_{2R}(x) = x$ and $\chi_R(x) = 1$, so

$$\widetilde{f}^{(R)}(t, x) = \widehat{f}_R\Big(t, \frac{x}{2R}\Big), \qquad x \in B_R.$$

Define the local loss density associated with the $H^1(\mu_t)$ norm by

$$\mathcal{L}(f, g) := |f - g|^2 + \|\nabla_x f - \nabla_x g\|_2^2, \tag{15}$$

where all terms are evaluated pointwise. We decompose the time-averaged global error as

$$\int_{t_0}^T \|\widetilde{f}^{(R)}_t - f^\star_t\|^2_{H^1(\mu_t)}\, dt = \underbrace{\int_{t_0}^T \int_{B_R} \mathcal{L}(\widetilde{f}^{(R)}, f^\star)\, d\mu_t\, dt}_{\textbf{Main error}} + \underbrace{\int_{t_0}^T \int_{B_R^c} \mathcal{L}(\widetilde{f}^{(R)}, f^\star)\, d\mu_t\, dt}_{\textbf{Tail error}}. \tag{16}$$

The main error is controlled by applying the compact-domain analysis to the pulled-back estimator on $C_{2R}$. The tail error is controlled by the sub-Gaussian decay of $\mu_t$ under Assumption 5.1, together with the growth bounds for $f^\star$ implied by the VP structure. Choosing $R = R(n)$ then balances these two errors and yields Theorem 5.2.

**Theorem 5.2** (Convergence for VP diffusion models on unbounded domains)**.** *Suppose Assumptions 5.1 hold. Let*

$$D := d+1, \qquad \alpha := \frac{s-1}{D + 2s - 2}.$$

*For each $R > 0$, consider the $R$-dependent clipped and spatial-gradient-bounded time-space class*

$$\mathcal{F}_R = \mathcal{F}^{(d+1)}_{M_R}(L, W, S, B), \qquad M_R = C_M(1+R).$$

*Take*

$$L = O(1), \qquad W, S, B = O\Big(n^{\frac{D}{D+2s-2}}\Big), \qquad \lambda \asymp n^{-\alpha}.$$

*Choose*

$$R = A\sqrt{\log n},$$

*where $A > 0$ is sufficiently large depending only on $t_0, T, R_{\mathrm{init}}, \overline{\alpha}, \underline{\sigma}, \overline{\sigma}$. Then, with probability at least $1 - 3n^{-2}$,*

$$\int_{t_0}^T \|\widetilde{f}^{(R)}_t - f^\star_t\|^2_{H^1(\mu_t)}\, dt \le Ce^{C\sqrt{\log n}}(\log n)^K n^{-\alpha},$$

*where $C, K < \infty$ are independent of $n$. Consequently, for every $\varepsilon > 0$, there exists $C_\varepsilon < \infty$ such that*

$$\int_{t_0}^T \|\widetilde{f}^{(R)}_t - f^\star_t\|^2_{H^1(\mu_t)}\, dt \le C_\varepsilon n^{-\alpha+\varepsilon}.$$

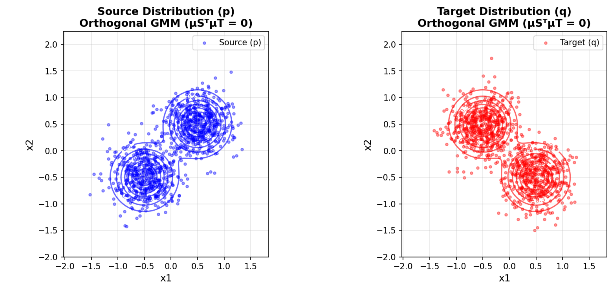


*Figure 3.* One example of the simulation distributions (All distribution pairs can be found in Appendix D.1).

# 6. Experiments

This section presents experiments on both synthetic environments and real-world datasets, demonstrating the promising performance of our proposed method.

## 6.1. Estimation error in simulation environment

In this simulation study, we show the effectiveness of our regularization, in terms of the accuracy of estimating the gradient of the log density ratio $\nabla \log \frac{q}{p}$.

We construct three distinct pairs of two–dimensional source and target distributions $(p, q)$, for which the quantity $\nabla \log \frac{q}{p}$ can be computed in closed form. Detailed descriptions of the three distribution pairs are provided in Appendix D.1. For each pair we train a three–layer MLP $f_\theta(x)$ to approximate the log density ratio.

We consider two training objectives: a standard classification-based objective, and our Sobolev-penalized classification objective. For each $(p, q)$ pair we train the network with $N \in \{10, 100, 1000\}$ samples drawn independently from both $p$ and $q$. After training, we set the gradient estimator directly as $g_\theta(x) := \nabla_x f_\theta(x)$, and estimate its mean squared error using $1000$ new samples from both $p$ and $q$.

The errors are reported in Table 2. We first observe that the Sobolev-penalized estimator (*Cls w sob*) consistently outperforms the naive classification baseline (*Cls w/o sob*) across all training sizes. Notably, this relative improvement is most pronounced in the small-sample regime (*e.g.*, more than a $50\%$ improvement for $N = 20$), confirming that the Sobolev penalty effectively stabilizes the estimator when data are scarce.

## 6.2. Transfer learning with real-world datasets

In transfer learning, the goal is to leverage labeled samples from a source distribution to improve prediction on a related target distribution. We observe source data $\mathcal{D}_p := \{(x_p^{(i)}, z_p^{(i)})\}_{i=1}^{n_p}$ drawn from a joint distribution $P$ and target samples $\mathcal{D}_q := \{(x_q^{(j)}, z_q^{(j)})\}_{j=1}^{n_q}$ drawn from a different joint distribution $Q$. In practice, labeled target samples are often scarce, making direct training unreliable. Our objective is therefore to infer a labeling function for $\mathcal{D}_q$ by transferring information from $\mathcal{D}_p$.

### 6.2.1. WGF METHODS

When many unlabeled target samples are available, following prior optimal-transport-based adaptation methods(Liu et al., 2023), we use WGF to transport labeled source particles toward the target distribution and then train a classifier on the transported samples.

WGF requires the score difference $\nabla \log q - \nabla \log p_t$. We therefore evaluate the proposed estimators by implementing WGF on the Office–Caltech-10 benchmark (Saenko et al., 2010), which contains four visual datasets: Amazon, Caltech, DSLR, and Webcam. Following standard practice, all samples are projected onto a 100-dimensional PCA subspace. We compare target classification accuracy across four settings: a source-only RBF SVM baseline, and RBF SVMs trained on WGF-transported source samples using (i) kernel-based estimators (LL; (Liu et al., 2023)), (ii) unregularized classification-based estimators, and (iii) our Sobolev-regularized classification estimator.

The results (Table 3) show that directly reusing source classifiers can lead to severe performance degradation (e.g., Amazon $\to$ DSLR), whereas joint distribution–based adaptation substantially alleviates this issue. Moreover, Sobolev regularization enables the classification-based method to achieve the strongest overall performance.

While kernel-based estimators attain accuracy comparable to our Sobolev-regularized approach in low-dimensional settings, they incur substantially higher computational cost. The kernel-based WGF baseline is intrinsically local, requiring a new gradient estimator to be trained at each WGF iteration through kernel optimization, which repeatedly evaluates interactions with the training samples (see Table 4). In contrast, the classification-based estimator amortizes gradient evaluation: once trained, the score difference at a new point is obtained via a single forward and backward pass of a fixed network. This amortization accounts for the significant speedup and enables scalability to large-scale settings, including diffusion-based generation.

### 6.2.2. DIFFUSION MODELS

When the amount of unlabeled target data is extremely limited, directly training a generative model on the target task is often unreliable. A more effective alternative is to leverage a large source domain and adapt a pre-trained diffusion model to the target distribution via transfer. In particular, recent work (Ouyang et al., 2024) proposes to guide a source-

*Table 1.* Accuracy for gradient of logdensity ratio estimation. Best results are marked in bold. Improvement indicates the relative reduction in error from Cls w/o sob to Cls w sob.

| Train Size | Cls w/o sob (A) | Cls w sob (B) | Kernel | Improvement $(A-B)/A$ [%] |
|---|---|---|---|---|
| **ROTATED_RIDGE** | | | | |
| 20 | $61.94 \pm 7.98$ | $30.96 \pm 6.92$ | $\mathbf{7.65 \pm 1.24}$ | 50.02 |
| 200 | $5.91 \pm 0.92$ | $\mathbf{4.29 \pm 0.61}$ | $6.44 \pm 0.43$ | 27.36 |
| 2000 | $1.68 \pm 0.21$ | $\mathbf{0.96 \pm 0.12}$ | $5.80 \pm 0.06$ | 42.96 |
| **ORTHOGONAL_GMM** | | | | |
| 20 | $24.10 \pm 2.73$ | $\mathbf{11.89 \pm 0.80}$ | $25.87 \pm 1.42$ | 50.66 |
| 200 | $8.90 \pm 2.74$ | $\mathbf{6.31 \pm 0.44}$ | $23.78 \pm 0.63$ | 29.19 |
| 2000 | $5.03 \pm 0.34$ | $\mathbf{3.83 \pm 0.17}$ | $22.80 \pm 0.21$ | 23.86 |
| **BOUNDED** | | | | |
| 20 | $103.46 \pm 7.16$ | $48.77 \pm 2.62$ | $\mathbf{7.73 \pm 0.59}$ | 52.86 |
| 200 | $15.68 \pm 2.45$ | $9.86 \pm 1.19$ | $\mathbf{3.68 \pm 0.09}$ | 37.12 |
| 2000 | $2.42 \pm 0.17$ | $\mathbf{1.96 \pm 0.13}$ | $3.55 \pm 0.01$ | 18.88 |

*Table 2.* Accuracy for gradient of log-density ratio estimation. Values are reported as mean (standard deviation). Best results are marked in bold.

| Train Size | Cls w/o sob (A) | Cls w sob (B) | Improve $(A-B)/A$ [%] |
|---|---|---|---|
| **ROTATED_RIDGE** | | | |
| 20 | 61.94(7.98) | $\mathbf{30.96(6.92)}$ | 50.02 |
| 200 | 5.91(0.92) | $\mathbf{4.29(0.61)}$ | 27.36 |
| 2000 | 1.68(0.21) | $\mathbf{0.96(0.12)}$ | 42.96 |
| **ORTHOGONAL_GMM** | | | |
| 20 | 24.10(2.73) | $\mathbf{11.89(0.80)}$ | 50.66 |
| 200 | 8.90(2.74) | $\mathbf{6.31(0.44)}$ | 29.19 |
| 2000 | 5.03(0.34) | $\mathbf{3.83(0.17)}$ | 23.86 |
| **BOUNDED** | | | |
| 20 | 103.46(7.16) | $\mathbf{48.77(2.62)}$ | 52.86 |
| 200 | 15.68(2.45) | $\mathbf{9.86(1.19)}$ | 37.12 |
| 2000 | 2.42(0.17) | $\mathbf{1.96(0.13)}$ | 18.88 |

*Table 3.* Comparison of classification accuracy on Office-Caltech-10 Dataset

| $\mathcal{D}_p \to \mathcal{D}_q$ | **Base** | **Cls w/o sob** | **Kernel** | **Cls w sob** |
|---|---|---|---|---|
| amz.→cal. | 0.7115 | 0.7863 | 0.8379 | **0.8504** |
| amz.→dslr | 0.2675 | 0.7580 | **0.7962** | **0.7962** |
| amz.→web. | 0.3932 | 0.7390 | **0.8678** | 0.8203 |
| cal.→amz. | **0.9081** | 0.8716 | **0.9081** | 0.8977 |
| cal.→dslr | 0.2420 | 0.8535 | 0.8344 | **0.8623** |
| cal.→web. | 0.3797 | 0.7763 | **0.8203** | 0.8034 |
| dslr→amz. | 0.7035 | 0.8017 | **0.8716** | 0.8006 |
| dslr→cal. | 0.6572 | 0.7489 | **0.8094** | 0.7427 |
| dslr→web. | 0.9492 | 0.9492 | 0.9492 | **0.9525** |
| web.→amz. | 0.6294 | 0.6002 | 0.5877 | **0.7046** |
| web.→cal. | 0.3954 | 0.7070 | **0.7640** | 0.6456 |
| web.→dslr | 0.8535 | 0.9618 | 0.9554 | **0.9682** |

*Table 4.* Average gradient computation time across the 12 transfer tasks shown in Table 3

| **Method** | **Gradient Calculation Time** |
|---|---|
| Baseline | N/A |
| Kernel | 2.0617 s |
| Cls w/o sob | **0.0004 s** |
| Cls w sob | **0.0004 s** |

trained diffusion model using an estimated score difference between the source and target distributions.

To evaluate the effectiveness of such transfer-based diffusion model methods under limited target data, we consider a benchmark task in electrocardiogram (ECG) generation. Following (Ouyang et al., 2024), we use the PTB-XL dataset (Wagner et al., 2020) as the source task and the ICBEB2018 dataset (Liu et al., 2018) as the target task, whose details can be found on Appendix D.3.

**Evaluation protocol.** We assess different methods through their ability to generate target-task samples for downstream classification. (More results on computing time and generation quality can be found on Appendix D.3) Specifically, each method is used to generate a sufficient number of synthetic ECG samples, which are then combined with the limited target samples to train a classifier. Performance is evaluated on the target test set. We compare four approaches: (1) *Vanilla Diffusion*, which trains a diffusion model directly on limited target data; (2) *Finetune Generator*, which adapts a source-trained diffusion model to generate target-label samples; (3) *TGDP* (Ouyang et al., 2024), which trains the guidance function using standard classification objectives; and (4) *TGDP-SoB*, which incorporates Sobolev regularization into the TGDP framework.

Following the ECG benchmark protocol in (Strodthoff et al., 2020), we report the Macro-averaged area under the ROC curve (AUC), macro-averaged $F_\beta$-score with $\beta = 2$, and macro-averaged $G_\beta$-score with $\beta = 2$; where $F_\beta = \frac{(1+\beta^2)\cdot \mathrm{TP}}{(1+\beta^2)\cdot \mathrm{TP}+\beta^2\cdot \mathrm{FN}+\mathrm{FP}}$, $G_\beta = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\beta\cdot \mathrm{FN}}$. As shown in Table 5, TGDP substantially outperforms the baseline methods across all evaluation metrics. In addition, the Sobolev-regularized variant (TGDP-SoB) consistently yields further improvements, highlighting the benefit of regularized score difference estimation for diffusion-based transfer under limited target data.

| **Method** | **AUC** | $\mathbf{F_{\beta=2}}$ | $\mathbf{G_{\beta=2}}$ |
|---|---|---|---|
| Vanilla Diffusion | 0.844(07) | 0.590(09) | 0.331(08) |
| Finetune Generator | 0.862(05) | 0.604(09) | 0.351(09) |
| TGDP | 0.905(04) | 0.662(10) | 0.436(12) |
| **TGDP-SoB** | **0.915(05)** | **0.693(11)** | **0.453(11)** |

*Table 5.* Results on ECG benchmark for downstream classification task. (90% confidence intervals are provided via empirical bootstrapping (Strodthoff et al., 2020); 0.915(04) stands for 0.915 ± 0.004.)

# 7. Broader Impact

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

# References


Bauer, F., Pereverzev, S., and Rosasco, L. On regularization algorithms in learning theory. *Journal of Complexity*, 23 (1):52–72, 2007.

Black, K., Janner, M., Du, Y., Kostrikov, I., and Levine, S. Training diffusion models with reinforcement learning. *arXiv preprint arXiv:2305.13301*, 2023.

Cong, D. N., Bao, H. T., and Hoang-Thanh, T. Guiding noisy label conditional diffusion models with score-based discriminator correction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 18531–18541, 2025.

Courty, N., Flamary, R., Tuia, D., and Rakotomamonjy, A. Optimal transport for domain adaptation. *IEEE transactions on pattern analysis and machine intelligence*, 39(9): 1853–1865, 2016.

Denker, A., Vargas, F., Padhy, S., Didi, K., Mathis, S. V., Barbano, R., Dutordoir, V., Mathieu, E., Komorowska, U. J., and Lio, P. Deft: Efficient fine-tuning of diffusion models by learning the generalised $h$-transform. In *Neurips*, volume 37, 2024.

Ding, Z., Duan, C., Jiao, Y., and Yang, J. Z. Semi-supervised deep sobolev regression: Estimation and variable selection by requ neural network. *IEEE Transactions on Information Theory*, 2025.

Du, Y., Plainer, M., Brekelmans, R., Duan, C., Noé, F., Gomes, C. P., Aspuru-Guzik, A., and Neklyudov, K. Doob's lagrangian: A sample-efficient variational approach to transition path sampling. In *Neurips*, volume 37, 2024.

Engl, H. W., Hanke, M., and Neubauer, A. *Regularization of Inverse Problems*. Kluwer Academic Publishers, 1996.

Fan, Y., Watkins, O., Du, Y., Liu, H., Ryu, M., Boutilier, C., Abbeel, P., Ghavamzadeh, M., Lee, K., and Lee, K. Reinforcement learning for fine-tuning text-to-image diffusion models. In *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS) 2023*. Neural Information Processing Systems Foundation, 2023.

Farrell, M. H., Liang, T., and Misra, S. Deep neural networks for estimation and inference. *Econometrica*, 89(1): 181–213, 2021.

Han, Y., Razaviyayn, M., and Xu, R. Neural network-based score estimation in diffusion models: Optimization and generalization. *arXiv preprint arXiv:2401.15604*, 2024a.

Han, Y., Razaviyayn, M., and Xu, R. Stochastic control for fine-tuning diffusion models: Optimality, regularity, and convergence. *arXiv preprint arXiv:2412.18164*, 2024b.

Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., and Hochreiter, S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.

Howard, S., Nüsken, N., and Pidstrigach, J. Control consistency losses for diffusion bridges. *arXiv preprint arXiv:2512.05070*, 2025.

Husain, H. and Nock, R. Generalization for discriminator-guided diffusion models via strong duality.

Kim, D., Kim, Y., Kwon, S. J., Kang, W., and Moon, I.-C. Refining generative process with discriminator guidance in score-based diffusion models. *arXiv preprint arXiv:2211.17091*, 2022.

Kong, L., Du, Y., Mu, W., Neklyudov, K., De Bortoli, V., Wu, D., Wang, H., Ferber, A., Ma, Y.-A., Gomes, C. P., et al. Diffusion models as constrained samplers for optimization with unknown constraints. *arXiv preprint arXiv:2402.18012*, 2024.

Li, P., Li, Z., Zhang, H., and Bian, J. On the generalization properties of diffusion models. *Advances in Neural Information Processing Systems*, 36:2097–2127, 2023.

Lin, S., Xia, X., Ren, Y., Yang, C., Xiao, X., and Jiang, L. Diffusion adversarial post-training for one-step video generation. *arXiv preprint arXiv:2501.08316*, 2025.

Liu, F., Liu, C., Zhao, L., Zhang, X., Wu, X., Xu, X., Liu, Y., Ma, C., Wei, S., He, Z., et al. An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection. *Journal of Medical Imaging and Health Informatics*, 8(7):1368–1373, 2018.

Liu, S., Yu, J., Simons, J., Yi, M., and Beaumont, M. Minimizing $f$-divergences by interpolating velocity fields. *arXiv preprint arXiv:2305.15577*, 2023.

Lu, Y., Chen, H., Lu, J., Ying, L., and Blanchet, J. Machine learning for elliptic pdes: Fast rate generalization bound, neural scaling law and minimax optimality. *arXiv preprint arXiv:2110.06897*, 2021.

Mescheder, L., Geiger, A., and Nowozin, S. Which training methods for gans do actually converge? In *International conference on machine learning*, pp. 3481–3490. PMLR, 2018.

Ouyang, Y., Xie, L., Zha, H., and Cheng, G. Transfer learning for diffusion models. *Advances in Neural Information Processing Systems*, 37:136962–136989, 2024.

Pidstrigach, J., Baker, E., Domingo-Enrich, C., Deligiannidis, G., and Nüsken, N. Conditioning diffusions using malliavin calculus. *arXiv preprint arXiv:2504.03461*, 2025.

Roth, K., Lucchi, A., Nowozin, S., and Hofmann, T. Stabilizing training of generative adversarial networks through regularization. *Advances in neural information processing systems*, 30, 2017.

Saenko, K., Kulis, B., Fritz, M., and Darrell, T. Adapting visual category models to new domains. In *European conference on computer vision*, pp. 213–226. Springer, 2010.

Schmidt-Hieber, J. Nonparametric regression using deep neural networks with relu activation function. 2020.

Srikanth, A., Asokan, S., Shetty, N., and Seelamantula, C. S. Accelerated diffusion using closed-form discriminator guidance.

Strodthoff, N., Wagner, P., Schaeffter, T., and Samek, W. Deep learning for ecg analysis: Benchmarks and insights from ptb-xl. *IEEE journal of biomedical and health informatics*, 25(5):1519–1528, 2020.

Suzuki, T. Adaptivity of deep relu network for learning in besov and mixed smooth besov spaces: optimal rate and curse of dimensionality. *arXiv preprint arXiv:1810.08033*, 2018.

Takezawa, K. *Introduction to nonparametric regression*. John Wiley & Sons, 2005.

Tang, W. and Zhou, F. Fine-tuning of diffusion models via stochastic control: entropy regularization and beyond. *arXiv preprint arXiv:2403.06279*, 2024.

Uehara, M., Zhao, Y., Black, K., Hajiramezanali, E., Scalia, G., Diamant, N. L., Tseng, A. M., Biancalani, T., and Levine, S. Fine-tuning of continuous-time diffusion models as entropy-regularized control. *arXiv preprint arXiv:2402.15194*, 2024.

Verine, A., Inane, A. M., Bronnec, F. L., Negrevergne, B., and Chevaleyre, Y. Improving discriminator guidance in diffusion models. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pp. 222–238. Springer, 2025.

Wagner, P., Strodthoff, N., Bousseljot, R.-D., Kreiseler, D., Lunze, F. I., Samek, W., and Schaeffter, T. Ptb-xl, a large publicly available electrocardiography dataset. *Scientific data*, 7(1):1–15, 2020.

Wang, X., Lin, B., Liu, D., Chen, Y.-C., and Xu, C. Bridging data gaps in diffusion models with adversarial noise-based transfer learning. In *Forty-first International Conference on Machine Learning*, 2024.

# A. Proof for Bounded Case (Section 3)

**Notation** Given the dataset $\mathcal{D}$, for any measurable function $\varphi$ we define its empirical expectation with respect to $\mathcal{D}$ by $\mathbb{E}_{\mathcal{D}}[\varphi(Y,X)] := \frac{1}{2n}\sum_{i=1}^{2n}\varphi(Y_i,X_i)$. Hence the empirical cross-entropy risk can be defined as

$$\widehat{L}_{\mathrm{CE},\mathcal{D}}(f) := \mathbb{E}_{\mathcal{D}}[\ell_{\mathrm{CE}}(Y_i, f(X_i))].$$

We first show that the minimizer for our energy functional exists and is unique.

**Lemma A.1** (Existence and uniqueness of the population risk minimizer in $\mathcal{H}_M$)**.** *For any regularization parameter $\lambda > 0$, the define the Sobolev-regularized population risk*

$$J_\lambda(f) := L_{CE}(f) + \lambda\mathcal{R}(f).$$

*Then $J_\lambda$ admits a unique minimizer $f_\lambda \in \mathcal{H}_M(\mu)$:*

$$f_\lambda := \underset{f\in\mathcal{H}_M(\mu)}{\arg\min}\ J_\lambda(f).$$

*Proof.* We proceed by establishing uniqueness via strong convexity and existence via the properties of the constraint set $\mathcal{H}_M(\mu)$.

From Lemma 3.3, the functional $J_\lambda$ satisfies the following strong convexity inequality for any $f, g \in \mathcal{H}_M(\mu)$:

$$J_\lambda(f) - J_\lambda(g) - DJ_\lambda(g)[f-g] \geq \frac{c_{min}}{2}\|f-g\|^2_{L^2(\mu)} + \lambda\|\nabla(f-g)\|^2_{L^2(\mu)}. \tag{17}$$

Since $\lambda > 0$ and $c_{min} > 0$, the right-hand side is strictly positive for any $f \neq g$ (in the $H^1(\mu)$ norm sense). Suppose there exist two distinct minimizers $f_1, f_2 \in \mathcal{H}_M(\mu)$ with $J_\lambda(f_1) = J_\lambda(f_2) = \inf_{f\in\mathcal{H}_M(\mu)} J_\lambda(f) = m$. By the strict convexity, for any $t \in (0,1)$, we have:

$$J_\lambda(tf_1 + (1-t)f_2) < tJ_\lambda(f_1) + (1-t)J_\lambda(f_2) = m.$$

This implies we have found an element with a risk strictly lower than the infimum $m$, which is a contradiction. Thus, the minimizer must be unique.

To apply the Direct Method on the constrained set, we first establish that the feasible set $\mathcal{H}_M(\mu)$ is a weakly closed subset of $H^1(\mu)$.

First, $\mathcal{H}_M(\mu)$ is convex. For any $f, g \in \mathcal{H}_M(\mu)$ and $t \in [0,1]$, by the triangle inequality:

$$\|tf + (1-t)g\|_{L^\infty} \leq t\|f\|_{L^\infty} + (1-t)\|g\|_{L^\infty} \leq tM + (1-t)M = M.$$

Thus, the convex combination remains in $\mathcal{H}_M(\mu)$. Second, $\mathcal{H}_M(\mu)$ is closed in the strong topology of $H^1(\mu)$. Let $\{f_n\} \subset \mathcal{H}_M(\mu)$ be a sequence converging to $f$ in $H^1(\mu)$. Convergence in $H^1$ implies convergence in $L^2$, which in turn implies the existence of a subsequence converging pointwise almost everywhere. Since $|f_n(x)| \leq M$ a.e., the pointwise limit must satisfy $|f(x)| \leq M$ a.e. Thus, $f \in \mathcal{H}_M(\mu)$. Since $\mathcal{H}_M(\mu)$ is a closed and convex subset of a Banach space, it is weakly closed.

Next we show coercivity. Although functions in $\mathcal{H}_M(\mu)$ are bounded in $L^\infty$, they are not a priori bounded in $H^1(\mu)$ (as gradients can be arbitrarily large). However, fixing a reference $g = 0 \in \mathcal{H}_M(\mu)$, the inequality (17) implies:

$$J_\lambda(f) \geq J_\lambda(0) + DJ_\lambda(0)[f] + C(\lambda)\|f\|^2_{H^1(\mu)}.$$

Since the quadratic term dominates the linear functional as $\|f\|_{H^1} \to \infty$, $J_\lambda$ is coercive. This ensures that any minimizing sequence $\{f_n\}_{n=1}^\infty \subset \mathcal{H}_M(\mu)$ such that $J_\lambda(f_n) \to \inf_{f\in\mathcal{H}_M(\mu)} J_\lambda(f)$ is bounded in the $H^1(\mu)$ norm.

Finaly, we take limit via weak lower semicontinuity. Since the minimizing sequence $\{f_n\} \subset \mathcal{H}_M(\mu)$ is bounded in the reflexive space $H^1(\mu)$, by the Banach-Alaoglu theorem, there exists a subsequence $\{f_{n_k}\}$ that converges weakly to some limit $f_\lambda \in H^1(\mu)$. Crucially, because $\mathcal{H}_M(\mu)$ is weakly closed, the limit must satisfy the constraints: $f_\lambda \in \mathcal{H}_M(\mu)$. Finally, since $J_\lambda$ is continuous and convex, it is weakly lower semicontinuous. Therefore:

$$J_\lambda(f_\lambda) \leq \liminf_{k\to\infty} J_\lambda(f_{n_k}) = \inf_{f\in\mathcal{H}_M(\mu)} J_\lambda(f).$$

Thus, the minimum is attained by $f_\lambda$ within the set $\mathcal{H}_M(\mu)$. □

Then we provide some properties about sparse neural network $\mathcal{F}$.

**Proposition A.2** (Boundedness of DNN). *The gradients and function value of $f \in \mathcal{F}$ are uniformly bounded: $\exists M \geq M^\star$, s.t*

$$\max\left\{ \sup_{f\in\mathcal{F}(\Omega)} \|f\|_{L^\infty(\mu)}, \sup_{f\in\mathcal{F}(\Omega)} \|\nabla f\|_{L^\infty(\mu)}, \|f^\star\|_{L^\infty(\mu)}, \|\nabla f^\star\|_{L^\infty(\mu)}, \right\} \leq M.$$

*Proof.* Proof can be found on the proof of Lemma A.15 and Lemma A.19 in (Lu et al., 2021). □

**Lemma A.3** (Tensor-product B-spline quasi-interpolation). *Fix integers $k, l \in \mathbb{N}$. Let*

$$0 = t_0^{(l)} < t_1^{(l)} < \cdots < t_l^{(l)} = 1, \qquad t_i^{(l)} := \frac{i}{l},$$

*be the uniform partition of $[0,1]$. Define the extended knot sequence*

$$t_{-k+1}^{(l)} = \cdots = t_{-1}^{(l)} = t_0^{(l)} = 0, \qquad t_l^{(l)} = t_{l+1}^{(l)} = \cdots = t_{l+k-1}^{(l)} = 1.$$

*Let*

$$I_{l,k} := \{-k+1, -k+2, \ldots, l-1\}.$$

*For $i \in I_{l,k}$, define the univariate B-spline of order $k$ by*

$$N_{l,i}^{(k)}(x) := (-1)^k \big(t_{i+k}^{(l)} - t_i^{(l)}\big) \big[t_i^{(l)}, t_{i+1}^{(l)}, \ldots, t_{i+k}^{(l)}\big] \left\{(x-t)_+^{k-1}\right\}, \qquad x \in [0,1], \tag{18}$$

*where $[\cdot]$ denotes the divided-difference operator with respect to the variable $t$, and $(a)_+ := \max\{a, 0\}$.*

*Equivalently, for interior indices $0 \leq i \leq l-k+1$,*

$$N_{l,i}^{(k)}(x) = \frac{l^{k-1}}{(k-1)!} \sum_{j=0}^{k} (-1)^j \binom{k}{j} \left(x - \frac{i+j}{l}\right)_+^{k-1}. \tag{19}$$

*The boundary splines are defined by the repeated endpoint knots above.*

*For a multi-index*

$$\mathbf{i} = (i_1, \ldots, i_d) \in I_{l,k}^d,$$

*define the tensor-product B-spline*

$$N_{l,\mathbf{i}}^{(k)}(x) := \prod_{m=1}^{d} N_{l,i_m}^{(k)}(x_m), \qquad x = (x_1, \ldots, x_d) \in [0,1]^d. \tag{20}$$

*Let $\{\Lambda_\mathbf{i}\}_{\mathbf{i}\in I_{l,k}^d}$ be the standard local B-spline quasi-interpolation functionals associated with the tensor basis $\{N_{l,\mathbf{i}}^{(k)}\}_{\mathbf{i}\in I_{l,k}^d}$. Define the quasi-interpolation operator*

$$Q_{k,l}g := \sum_{\mathbf{i}\in I_{l,k}^d} \Lambda_\mathbf{i}(g)\, N_{l,\mathbf{i}}^{(k)}. \tag{21}$$

*Then $Q_{k,l}$ is local and stable. In particular, there exists a constant $C_{\mathrm{qi}} < \infty$, depending only on $k$ and $d$, but independent of $l$, such that for every $g \in W^{1,\infty}([0,1]^d)$,*

$$\|Q_{k,l}g\|_{L^\infty} \leq C_{\mathrm{qi}} \|g\|_{L^\infty}, \qquad \|\nabla Q_{k,l}g\|_{L^\infty} \leq C_{\mathrm{qi}} \|\nabla g\|_{L^\infty}. \tag{22}$$

*Moreover, for every $g \in H^s([0,1]^d)$, $s \in \mathbb{N}$, and every $0 \leq r \leq s$, if $k \geq s$, then*

$$\|Q_{k,l}g - g\|_{H^r} \leq C l^{-(s-r)} \|g\|_{H^s}, \tag{23}$$

*where $C$ depends only on $k, s, r, d$.*

*Proof.* The definitions (18)–(20) are the standard univariate and tensor-product B-splines with repeated endpoint knots. The operator $Q_{k,l}$ in (21) is the standard local B-spline quasi-interpolation operator.

We first recall its stability. Since the B-splines have compact support, form a partition of unity, and have uniformly bounded overlap, each point $x \in [0,1]^d$ belongs to the support of only $O_{k,d}(1)$ tensor-product basis functions. The local functionals $\Lambda_{\mathbf{i}}$ depend only on $g$ in a fixed-size neighborhood of $\operatorname{supp}(N_{l,\mathbf{i}}^{(k)})$, and satisfy the standard coefficient stability bounds

$$|\Lambda_{\mathbf{i}}(g)| \leq C_{k,d} \|g\|_{L^\infty(\Omega_{\mathbf{i}})},$$

where $\Omega_{\mathbf{i}}$ is the local support neighborhood associated with $N_{l,\mathbf{i}}^{(k)}$. Therefore,

$$\|Q_{k,l} g\|_{L^\infty} \leq C_{\mathrm{qi}} \|g\|_{L^\infty}.$$

For the derivative bound, differentiating

$$Q_{k,l} g = \sum_{\mathbf{i}} \Lambda_{\mathbf{i}}(g) N_{l,\mathbf{i}}^{(k)}$$

produces derivatives of the B-spline basis of size $O(l)$. The locality and polynomial-reproduction property of the quasi-interpolant imply the standard first-order coefficient-difference bound

$$|\Lambda_{\mathbf{i}}(g) - \Lambda_{\mathbf{j}}(g)| \leq C_{k,d} l^{-1} \|\nabla g\|_{L^\infty}$$

for neighboring indices $\mathbf{i}, \mathbf{j}$. The factor $l^{-1}$ in the coefficient difference cancels the $O(l)$ factor from differentiating the B-splines. Using again the uniformly bounded overlap of the tensor basis yields

$$\|\nabla Q_{k,l} g\|_{L^\infty} \leq C_{\mathrm{qi}} \|\nabla g\|_{L^\infty}.$$

This proves (22).

Finally, the approximation estimate (23) is the standard Sobolev approximation property of tensor-product B-spline quasi-interpolation. It follows from the Bramble–Hilbert lemma and the polynomial reproduction of $Q_{k,l}$ up to degree $k-1$. This gives

$$\|Q_{k,l} g - g\|_{H^r} \leq C l^{-(s-r)} \|g\|_{H^s},$$

whenever $k \geq s$ and $0 \leq r \leq s$. □

**Proposition A.4** (Approximation by the clipped and gradient-bounded sieve)**.** *Let $\mu_{p,q}$ be the mixture measure corresponding to a pair $(p,q)$ satisfying Assumption 2.1. Suppose $f^\star \in H^s(\mu_{p,q})$, $s \geq 2$, and*

$$\max\left\{\|f^\star\|_{L^\infty(\mu_{p,q})}, \|\nabla f^\star\|_{L^\infty(\mu_{p,q})}\right\} \leq M^\star.$$

*Let $C_{\mathrm{qi}}$ be the stability constant in Lemma A.3. Choose $M > C_{\mathrm{qi}} M^\star$, and choose the clipping map $T_M$ so that it is the identity on $[-M_0, M_0]$ for some*

$$C_{\mathrm{qi}} M^\star < M_0 < M.$$

*Then, for $L = O(1)$, $W = O(N)$, $S = O(N)$, and $B = O(N)$, there exists*

$$f_N^{\mathrm{app}} \in \mathcal{F}_M(L, W, S, B)$$

*such that*

$$\|f_N^{\mathrm{app}} - f^\star\|_{H^1(\mu_{p,q})}^2 \lesssim N^{-\frac{2(s-1)}{d}} \|f^\star\|_{H^s(\mu_{p,q})}^2. \tag{24}$$

*Proof.* It suffices to prove the result for $\Omega \subseteq [0,1]^d$. Under Assumption 2.1, the weighted Sobolev norm induced by $\mu_{p,q}$ is equivalent to the usual Sobolev norm on $\Omega$.

Let

$$l := \left\lceil N^{1/d} \right\rceil.$$

Following the ReLU$^3$ approximation construction, take the cubic B-spline quasi-interpolant

$$u_{\mathrm{sp}} := Q_{4,l} f^\star = \sum_{\mathbf{i} \in I_{l,4}^d} \Lambda_{\mathbf{i}}(f^\star) N_{l,\mathbf{i}}^{(4)}.$$

By Lemma A.3, with $k = 4$ and $r = 1$,

$$\|Q_{4,l} f^\star - f^\star\|_{H^1(\mu_{p,q})} \lesssim l^{-(s-1)} \|f^\star\|_{H^s(\mu_{p,q})}. \tag{25}$$

Since $l \asymp N^{1/d}$, this gives

$$\|Q_{4,l} f^\star - f^\star\|_{H^1(\mu_{p,q})} \lesssim N^{-\frac{s-1}{d}} \|f^\star\|_{H^s(\mu_{p,q})}.$$

We next verify that the spline approximant lies inside the fixed envelope. By the $W^{1,\infty}$-stability in Lemma A.3,

$$\|Q_{4,l} f^\star\|_{L^\infty(\mu_{p,q})} \le C_{\mathrm{qi}} \|f^\star\|_{L^\infty(\mu_{p,q})} \le C_{\mathrm{qi}} M^\star < M_0,$$

and

$$\|\nabla Q_{4,l} f^\star\|_{L^\infty(\mu_{p,q})} \le C_{\mathrm{qi}} \|\nabla f^\star\|_{L^\infty(\mu_{p,q})} \le C_{\mathrm{qi}} M^\star < M.$$

It remains to realize $Q_{4,l} f^\star$ as a sparse ReLU$^3$ neural network. By the explicit formula for the cubic B-splines,

$$N_{l,i}^{(4)}(x) = \frac{l^3}{3!} \sum_{j=0}^{4} (-1)^j \binom{4}{j} \left( x - \frac{i+j}{l} \right)_+^3$$

for interior indices, and by the analogous endpoint formulas for boundary splines, every univariate cubic B-spline can be implemented by a ReLU$^3$ subnetwork. Moreover, products of nonnegative spline factors can be implemented by ReLU$^3$ networks using the identity

$$xy = \frac{1}{2}\big[(x+y)^2 - x^2 - y^2\big],$$

together with the exact representation of $x^2$ by ReLU$^3$ units. Thus each tensor-product spline

$$N_{l,\mathbf{i}}^{(4)}(x) = \prod_{m=1}^{d} N_{l,i_m}^{(4)}(x_m)$$

can be implemented by a ReLU$^3$ subnetwork. Summing over $\mathbf{i} \in I_{l,4}^d$ with coefficients $\Lambda_{\mathbf{i}}(f^\star)$ gives a sparse ReLU$^3$ network $F_N$ satisfying

$$F_N = Q_{4,l} f^\star \quad \text{on } \Omega.$$

As in the standard spline implementation argument, this network can be chosen with

$$L = O(1), \qquad W = O(N), \qquad S = O(N), \qquad B = O(N).$$

Define

$$f_N^{\mathrm{app}} := T_M \circ F_N.$$

Since

$$\|F_N\|_{L^\infty(\mu_{p,q})} = \|Q_{4,l} f^\star\|_{L^\infty(\mu_{p,q})} < M_0,$$

and $T_M$ is the identity on $[-M_0, M_0]$, the clipping is inactive:

$$f_N^{\mathrm{app}} = F_N = Q_{4,l} f^\star.$$

Therefore

$$\|f_N^{\mathrm{app}}\|_{L^\infty(\mu_{p,q})} \le M, \qquad \|\nabla f_N^{\mathrm{app}}\|_{L^\infty(\mu_{p,q})} \le M,$$

so

$$f_N^{\mathrm{app}} \in \mathcal{F}_M(L, W, S, B).$$

Finally, (25) yields

$$\|f_N^{\mathrm{app}} - f^\star\|_{H^1(\mu_{p,q})} \lesssim N^{-\frac{s-1}{d}} \|f^\star\|_{H^s(\mu_{p,q})}.$$

Squaring both sides proves (24). □

### A.1. Proof of Lemma 3.3

*Proof.* We first analyze the cross-entropy loss. For any $x$, let $G_x(u) = -\eta(x)\log\sigma(u) - (1-\eta(x))\log(1-\sigma(u))$ be the conditional risk. A direct calculation yields the second derivative:

$$G''_x(u) = \sigma(u)(1-\sigma(u)) = \frac{1}{4\cosh^2(u/2)}.$$

Since any $f \in \mathcal{H}_M$ satisfies $|f(x)| \leq M$ almost everywhere, the curvature is uniformly bounded from below and above:

$$c_{\min} \;\leq\; G''_x(u) \;\leq\; \frac{1}{4}, \quad \forall u \in [-M, M]. \tag{26}$$

Since $G_x(u)$ is smooth with bounded derivatives on this compact interval, standard arguments from the calculus of variations imply that $L_{\mathrm{CE}}$ is Fréchet differentiable with $DL_{\mathrm{CE}}(g)[h] = \mathbb{E}[G'_X(g(X))h(X)]$. Applying Taylor's theorem with integral remainder to $G_x$, we obtain:

$$G_x(u) - G_x(v) - G'_x(v)(u-v) = (u-v)^2 \int_0^1 (1-t)G''_x\big(v + t(u-v)\big)\,\mathrm{d}t.$$

Using the bounds on $G''_x$, this implies

$$\frac{c_{\min}}{2}(u-v)^2 \;\leq\; G_x(u) - G_x(v) - G'_x(v)(u-v) \;\leq\; \frac{1}{8}(u-v)^2.$$

Integrating with respect to $\mu$ (setting $u = f(x), v = g(x)$) yields the strong convexity and smoothness estimates for $L_{\mathrm{CE}}$:

$$\frac{c_{\min}}{2}\|f-g\|^2_{L^2} \;\leq\; L_{\mathrm{CE}}(f) - L_{\mathrm{CE}}(g) - DL_{\mathrm{CE}}(g)[f-g] \;\leq\; \frac{1}{2}\|f-g\|^2_{L^2}. \tag{27}$$

We then analyze the regularizer. The functional $\mathcal{R}(f) = \|\nabla f\|^2_{L^2(\mu)}$ is a standard quadratic form on the Hilbert space $H^1(\mu)$. It is strictly convex and Fréchet differentiable with $D\Omega(g)[h] = 2\langle \nabla g, \nabla h\rangle_{L^2}$. The polarization identity yields the exact expansion:

$$\mathcal{R}(f) - \Omega(g) - D\Omega(g)[f-g] = \|\nabla(f-g)\|^2_{L^2(\mu)}. \tag{28}$$

Finally, we put together above results. Recall $J_\lambda(f) = L_{\mathrm{CE}}(f) + \lambda\mathcal{R}(f)$. By linearity, $DJ_\lambda = DL_{\mathrm{CE}} + \lambda D\Omega$. Combining the lower bound from (27) and the identity (28), we obtain:

$$J_\lambda(f) - J_\lambda(g) - DJ_\lambda(g)[f-g] \;\geq\; \frac{c_{\min}}{2}\|f-g\|^2_{L^2} + \lambda\|\nabla(f-g)\|^2_{L^2},$$

which proves (10). Similarly, combining the upper bound from (27) with (28) yields (10). □

### A.2. Proof of Lemma 3.4

*Proof.* Denote $\delta_\lambda := f_\lambda - f^\star$. We proceed in three steps.

Step 1: A pointwise calibration inequality along $f_\lambda - f^\star$. From (26) we have

$$c_{\min} \;\leq\; G''_x(u) \;\leq\; \frac{1}{4} \qquad \text{for all } u \in [-M, M] \text{ and all } x.$$

By the fundamental theorem of calculus,

$$G'_x(f(x)) - G'_x(f^\star(x)) = \int_0^1 G''_x\big(f^\star(x) + t(f(x) - f^\star(x))\big)\,(f(x) - f^\star(x))\,\mathrm{d}t.$$

Since $f^\star$ is the Bayes logit, we have $G'_x(f^\star(x)) = \sigma(f^\star(x)) - \eta(x) = 0$, so

$$G'_x(f(x))\,(f(x) - f^\star(x)) = \left(\int_0^1 G''_x\big(f^\star(x) + t(f(x) - f^\star(x))\big)\,\mathrm{d}t\right)(f(x) - f^\star(x))^2.$$

Define

$$w(x) := \int_0^1 G_x''\big(f^\star(x) + t(f(x) - f^\star(x))\big)\,\mathrm{d}t.$$

By the bounds on $G_x''$, we have $c_{\min} \le w(x) \le c_{\max}$ for all $x$. Moreover $G_x'(f(x)) = \sigma(f(x)) - \eta(x)$, so

$$(\sigma(f(x)) - \eta(x))\,(f(x) - f^\star(x)) = w(x)\,(f(x) - f^\star(x))^2.$$

Taking expectation with respect to $X \sim \mu$ gives

$$\mathbb{E}\big[(\sigma(f(X)) - \eta(X))\,(f(X) - f^\star(X))\big] = \mathbb{E}\big[w(X)\,(f(X) - f^\star(X))^2\big],$$

and hence

$$c_{\min}\,\|f - f^\star\|^2_{L^2(\mu)} \;\le\; \mathbb{E}\big[(\sigma(f(X)) - \eta(X))\,(f(X) - f^\star(X))\big] \;\le\; \frac{1}{4}\,\|f - f^\star\|^2_{L^2(\mu)}. \tag{29}$$

This is the desired "gradient-distance" inequality along the direction $f - f^\star$.

Step 2: First-order optimality of $f_\lambda$ and Green's identity. Since $f_\lambda$ minimizes $J_\lambda(f)$ over the convex set $\mathcal{H}_M$, and noting that $f^\star \in \mathcal{H}_M$ (which holds by the assumption $M \ge M^\star$), $f_\lambda$ satisfies the first-order variational inequality:

$$DJ_\lambda(f_\lambda)[f^\star - f_\lambda] \ge 0.$$

Let $\delta_\lambda := f_\lambda - f^\star$. The inequality above is equivalent to $DJ_\lambda(f_\lambda)[\delta_\lambda] \le 0$, which expands to:

$$\mathbb{E}\big[(\sigma(f_\lambda) - \eta)\delta_\lambda\big] + 2\lambda\,(\nabla f_\lambda, \nabla\delta_\lambda)_{L^2(\mu)} \;\le\; 0. \tag{30}$$

Decomposing $\nabla f_\lambda = \nabla f^\star + \nabla\delta_\lambda$ gives

$$(\nabla f_\lambda, \nabla\delta_\lambda)_{L^2(\mu)} = \|\nabla\delta_\lambda\|^2_{L^2(\mu)} + (\nabla f^\star, \nabla\delta_\lambda)_{L^2(\mu)}.$$

By the Neumann boundary condition in Assumption 2.1 and Green's Identity, we have

$$-(\nabla\delta_\lambda, \nabla f^\star)_{L^2(\mu)} = (\Delta f^\star + \nabla f^\star \cdot \nabla \log\mu,\ \delta_\lambda)_{L^2(\mu)}.$$

Recall

$$R_0^\star := \Delta f^\star + \nabla f^\star \cdot \nabla \log\mu \in L^2(\mu).$$

Then

$$(\nabla f^\star, \nabla\delta_\lambda)_{L^2(\mu)} = -(R_0^\star, \delta_\lambda)_{L^2(\mu)}.$$

Substituting into (30) yields

$$\mathbb{E}\big[(\sigma(f_\lambda) - \eta)\delta_\lambda\big] + 2\lambda\|\nabla\delta_\lambda\|^2_{L^2(\mu)} - 2\lambda(R_0^\star, \delta_\lambda)_{L^2(\mu)} \;\le\; 0,$$

or equivalently

$$\mathbb{E}\big[(\sigma(f_\lambda) - \eta)\delta_\lambda\big] + 2\lambda\|\nabla\delta_\lambda\|^2_{L^2(\mu)} \;\le\; 2\lambda(R_0^\star, \delta_\lambda)_{L^2(\mu)}. \tag{31}$$

Step 3: Combining the sandwich inequality and Cauchy–Schwarz. Applying the lower bound in (29) with $f = f_\lambda$, we obtain

$$c_{\min}\,\|\delta_\lambda\|^2_{L^2(\mu)} \;\le\; \mathbb{E}\big[(\sigma(f_\lambda) - \eta)\delta_\lambda\big].$$

Combining this with (31), we get

$$c_{\min}\,\|\delta_\lambda\|^2_{L^2(\mu)} + 2\lambda\|\nabla\delta_\lambda\|^2_{L^2(\mu)} \;\le\; 2\lambda(R_0^\star, \delta_\lambda)_{L^2(\mu)} \le 2\lambda\,\|R_0^\star\|_{L^2(\mu)}\,\|\delta_\lambda\|_{L^2(\mu)}. \tag{32}$$

We now extract the desired bounds from (32).

*(i) Value bias.* Dropping the nonnegative gradient term on the left-hand side gives

$$c_{\min}\,\|\delta_\lambda\|_{L^2(\mu)}^2 \le 2\lambda\,\|R_0^\star\|_{L^2(\mu)}\,\|\delta_\lambda\|_{L^2(\mu)}.$$

If $\delta_\lambda \equiv 0$, the conclusion is trivial. Otherwise, dividing both sides by $\|\delta_\lambda\|_{L^2(\mu)}$,

$$\|\delta_\lambda\|_{L^2(\mu)} \le \frac{2\,\|R_0^\star\|_{L^2(\mu)}}{c_{\min}}\,\lambda.$$

Hence

$$\|f_\lambda - f^\star\|_{L^2(\mu)}^2 = \|\delta_\lambda\|_{L^2(\mu)}^2 \le \Big(\frac{2\,\|R_0^\star\|_{L^2(\mu)}}{c_{\min}}\Big)^2\,\lambda^2. \tag{33}$$

*(ii) Gradient bias.* Substituting the bound $\|\delta_\lambda\|_{L^2(\mu)} \le \big(2\,\|R_0^\star\|_{L^2(\mu)}/c_{\min}\big)\lambda$ back into (32), we obtain

$$2\lambda\|\nabla\delta_\lambda\|_{L^2(\mu)}^2 \le 2\lambda\,\|R_0^\star\|_{L^2(\mu)}\,\|\delta_\lambda\|_{L^2(\mu)} \le 2\lambda\,\|R_0^\star\|_{L^2(\mu)}\cdot\frac{2\,\|R_0^\star\|_{L^2(\mu)}}{c_{\min}}\,\lambda = \frac{4\,\|R_0^\star\|_{L^2(\mu)}^2}{c_{\min}}\,\lambda^2.$$

Dividing by $2\lambda$ (recall $\lambda > 0$) yields

$$\|\nabla(f_\lambda - f^\star)\|_{L^2(\mu)}^2 = \|\nabla\delta_\lambda\|_{L^2(\mu)}^2 \le \frac{2\,\|R_0^\star\|_{L^2(\mu)}^2}{c_{\min}}\,\lambda. \tag{34}$$

Using the explicit value $c_{\min} = 1/(4\cosh^2(M/2))$, we can write

$$\Big(\frac{2\,\|R_0^\star\|_{L^2(\mu)}}{c_{\min}}\Big)^2 = 64\,\|R_0^\star\|_{L^2(\mu)}^2\,\cosh^4(M/2), \qquad \frac{2\,\|R_0^\star\|_{L^2(\mu)}^2}{c_{\min}} = 8\,\|R_0^\star\|_{L^2(\mu)}^2\,\cosh^2(M/2).$$

Thus we may take $\beta = 64\,\|R_0^\star\|_{L^2(\mu)}^2\,\cosh^4\big(\tfrac{M}{2}\big)$ so that

$$\max\Big\{64\,\|R_0^\star\|_{L^2(\mu)}^2\,\cosh^4\big(\tfrac{M}{2}\big),\ 8\,\|R_0^\star\|_{L^2(\mu)}^2\,\cosh^2\big(\tfrac{M}{2}\big)\Big\} \le \beta.$$

□

### A.3. Proof of Theorem 3.5

Let $\{\sigma_i\}_{i=1}^{2n}$ be i.i.d. Rademacher random variables, i.e. $\mathbb{P}(\sigma_i = 1) = \mathbb{P}(\sigma_i = -1) = 1/2$. The empirical Rademacher complexity of a function class $\mathcal{G}$ with respect to $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^{2n}$ is defined as

$$R_n(\mathcal{G}) := \mathbb{E}_\sigma\left[\sup_{g\in\mathcal{G}} \frac{1}{2n}\sum_{i=1}^{2n} \sigma_i g(X_i, Y_i)\,\middle|\,\mathcal{D}\right].$$

*Proof.* Fix $t > 0$. For $f \in \mathcal{F}$, define

$$E(f) := J_\lambda(f) - J_\lambda(f_\lambda) \ge 0.$$

Define the shifted excess-loss function

$$g_f(x,y) := \ell_{\mathrm{CE}}(y, f(x)) - \ell_{\mathrm{CE}}(y, f_0(x)) + \lambda\left(\|\nabla f(x)\|_2^2 - \|\nabla f_0(x)\|_2^2\right).$$

Then

$$\mathbb{E}[g_f] = J_\lambda(f) - J_\lambda(f_0) = E(f) - E(f_0),$$

and

$$\frac{1}{2n}\sum_{i=1}^{2n} g_f(X_i, Y_i) = \widehat{J}_{\lambda,\mathcal{D}}(f) - \widehat{J}_{\lambda,\mathcal{D}}(f_0).$$

Step 1: Local Rademacher complexity bound. For $r > 0$, define the localized set

$$L_r^{\mathrm{sh}}(\Omega) := \{f \in \mathcal{F} : E(f) + E(f_0) \le r\},$$

and define the shifted localized loss class

$$\mathcal{S}_r^{\mathrm{sh}}(\Omega) := \left\{g_f : \Omega \times \{0,1\} \to \mathbb{R} \;\middle|\; f \in L_r^{\mathrm{sh}}(\Omega)\right\}.$$

We claim that for $r \gtrsim n^{-2}$,

$$R_n(\mathcal{S}_r^{\mathrm{sh}}(\Omega)) \le \phi(r) := C_M \sqrt{\frac{S3^L r}{n} \log(BWn)}, \tag{35}$$

and $\phi(4r) \le 2\phi(r)$.

We first prove the Lipschitz property of the shifted excess loss. Since

$$\partial_u \ell_{\mathrm{CE}}(y,u) = \sigma(u) - y,$$

the cross-entropy loss is 1-Lipschitz in $u$. For any $f_1, f_2 \in L_r^{\mathrm{sh}}(\Omega)$, using the fixed clipped-gradient envelope

$$\|f_j\|_{L^\infty(\mu)} \le M, \qquad \|\nabla f_j\|_{L^\infty(\mu)} \le M, \qquad j = 1,2,$$

we obtain

$$\begin{aligned} |g_{f_1}(x,y) - g_{f_2}(x,y)| &= \left|\ell_{\mathrm{CE}}(y, f_1(x)) - \ell_{\mathrm{CE}}(y, f_2(x)) + \lambda\left(\|\nabla f_1(x)\|_2^2 - \|\nabla f_2(x)\|_2^2\right)\right| \\ &\le |f_1(x) - f_2(x)| + \lambda\left(\|\nabla f_1(x)\|_2 + \|\nabla f_2(x)\|_2\right)\|\nabla f_1(x) - \nabla f_2(x)\|_2 \\ &\le |f_1(x) - f_2(x)| + 2M\lambda\|\nabla f_1(x) - \nabla f_2(x)\|_2. \end{aligned}$$

Thus $g_f$ is Lipschitz in $(f, \lambda\nabla f)$, with Lipschitz constant depending only on $M$.

Next we relate the shifted localization $E(f) + E(f_0) \le r$ to a radius constraint around $f_0$. By Lemma 3.3, for all $f \in \mathcal{H}_M$,

$$E(f) = J_\lambda(f) - J_\lambda(f_\lambda) \ge \frac{c_{\min}}{2}\|f - f_\lambda\|_{L^2(\mu)}^2 + \lambda\|\nabla(f - f_\lambda)\|_{L^2(\mu)}^2.$$

Similarly,

$$E(f_0) = J_\lambda(f_0) - J_\lambda(f_\lambda) \ge \frac{c_{\min}}{2}\|f_0 - f_\lambda\|_{L^2(\mu)}^2 + \lambda\|\nabla(f_0 - f_\lambda)\|_{L^2(\mu)}^2.$$

Therefore, if $f \in L_r^{\mathrm{sh}}(\Omega)$, then

$$\|f - f_0\|_{L^2(\mu)}^2 \le 2\|f - f_\lambda\|_{L^2(\mu)}^2 + 2\|f_0 - f_\lambda\|_{L^2(\mu)}^2 \le \frac{4}{c_{\min}} r,$$

and

$$\|\lambda\nabla(f - f_0)\|_{L^2(\mu)}^2 \le 2\lambda^2\|\nabla(f - f_\lambda)\|_{L^2(\mu)}^2 + 2\lambda^2\|\nabla(f_0 - f_\lambda)\|_{L^2(\mu)}^2 \le 2\lambda r \le 2r,$$

where we used $0 < \lambda < 1$.

Applying the Talagrand contraction lemma to the Lipschitz inequality above, and then applying the derivative-augmented covering-number argument as in Lemma A.26 of (Lu et al., 2021), we obtain, for all $r \gtrsim n^{-2}$,

$$\begin{aligned} \mathfrak{R}_n(\mathcal{S}_r^{\mathrm{sh}}(\Omega)) &\overset{\diamond_1}{\lesssim_M} \mathfrak{R}_n\big(\{f - f_0 : f \in \mathcal{L}_r^{\mathrm{sh}}(\Omega)\}\big) \\ &\qquad + \mathfrak{R}_n\big(\{\lambda\nabla f - \lambda\nabla f_0 : f \in \mathcal{L}_r^{\mathrm{sh}}(\Omega)\}\big) \\ &\overset{\diamond_2}{\lesssim_M} \frac{1}{n} + \frac{1}{\sqrt{n}}\int_{1/n}^{C\sqrt{r}} \sqrt{S\Big[\log(\delta^{-1}) + 3L\log(WB)\Big]}\, d\delta \\ &\overset{\diamond_3}{\lesssim_M} \sqrt{\frac{S^3 L r}{n}\log(BWn)}. \end{aligned} \tag{A.xx}$$

Here $\diamond_1$ follows from the Talagrand contraction lemma applied to the Lipschitz bound

$$|g_{f_1}(x,y) - g_{f_2}(x,y)| \le |f_1(x) - f_2(x)| + 2M\lambda \|\nabla f_1(x) - \nabla f_2(x)\|_2.$$

Thus the Rademacher complexity of the shifted loss class is reduced to the complexities of the localized function and gradient classes.

The step $\diamond_2$ uses the shifted localization $E(f) + E(f_0) \le r$. By strong convexity of $J_\lambda$,

$$\|f - f_0\|^2_{L^2(\mu)} + \|\lambda \nabla (f - f_0)\|^2_{L^2(\mu)} \lesssim_M r.$$

Hence both localized classes have $L^2$-radius of order $\sqrt{r}$. Combining this radius bound with the derivative-augmented covering-number estimate for sparse ReLU$^3$ networks yields the Dudley entropy integral with upper limit $C\sqrt{r}$.

Finally, $\diamond_3$ evaluates the Dudley integral and absorbs the $1/n$ term into the square-root term using $r \gtrsim n^{-2}$. Therefore

$$\mathfrak{R}_n(\mathcal{S}_r^{\mathrm{sh}}(\Omega)) \le \phi(r) := C_M \sqrt{\frac{S^3 L r}{n} \log(BWn)}.$$

The sub-root property $\phi(4r) \le 2\phi(r)$ follows immediately from the square-root dependence on $r$.

Step 2: Peeling and the normalized empirical process. The local Rademacher bound in Step 1 controls the class only on each localized shell

$$\mathcal{S}_s^{\mathrm{sh}}(\Omega) = \{g_f : E(f) + E(f_0) \le s\}.$$

To turn these shell-wise bounds into a uniform bound over the whole class $\mathcal{F}$, we introduce the normalized class

$$\widehat{\mathcal{S}}_r^{\mathrm{sh}}(\Omega) := \left\{ \widehat{g}_f : \widehat{g}_f(x,y) = \frac{g_f(x,y)}{E(f) + E(f_0) + r},\ f \in \mathcal{F} \right\}.$$

The denominator is positive because $E(f) \ge 0$, $E(f_0) \ge 0$, and $r > 0$. The role of the denominator is to rescale each function according to its own local radius $E(f) + E(f_0)$. Thus, functions lying farther from the population minimizer are placed in larger shells and are normalized more strongly.

Applying the peeling lemma, Lemma A.7 in (Lu et al., 2021), with the localization functional

$$U(f) := E(f) + E(f_0),$$

and using the sub-root bound

$$R_n(\mathcal{S}_s^{\mathrm{sh}}(\Omega)) \le \phi(s),$$

we obtain

$$R_n(\widehat{\mathcal{S}}_r^{\mathrm{sh}}(\Omega)) \le \frac{4\phi(r)}{r}.$$

In words, peeling converts the local Rademacher bounds on the shells $\{U(f) \le s\}$ into a single Rademacher bound for the normalized global class.

We now pass from the normalized loss class to the centered normalized empirical process. Define

$$\widetilde{\mathcal{T}}_r^{\mathrm{sh}}(\Omega) := \left\{ \widetilde{g}_f : \Omega \times \{0,1\} \to \mathbb{R} \;\middle|\; \widetilde{g}_f(x,y) = \frac{\mathbb{E}[g_f] - g_f(x,y)}{E(f) + E(f_0) + r},\ f \in \mathcal{F} \right\}.$$

Let $(X_i', Y_i')_{i=1}^{2n}$ be an independent ghost sample. By the symmetrization lemma, Lemma A.3 in (Lu et al., 2021),

$$\begin{aligned} \sup_{\widetilde{g} \in \widetilde{\mathcal{T}}_r^{\mathrm{sh}}} \mathbb{E}\left[ \frac{1}{2n} \sum_{i=1}^{2n} \widetilde{g}(X_i', Y_i') \right] &\le 2R_n(\widehat{\mathcal{S}}_r^{\mathrm{sh}}(\Omega)) \\ &\le \frac{8\phi(r)}{r}. \end{aligned}$$

Therefore,

$$\sup_{\widetilde{g}\in\widetilde{\mathcal{T}}_r^{\mathrm{sh}}} \mathbb{E}\left[\frac{1}{2n}\sum_{i=1}^{2n}\widetilde{g}(X_i',Y_i')\right] \le \frac{8\phi(r)}{r}. \tag{36}$$

Step 3: Verifying the Talagrand conditions. For any $f \in \mathcal{F}$, since both $f$ and $f_0$ lie in the clipped and gradient-bounded sieve,

$$\|f\|_\infty, \|f_0\|_\infty, \|\nabla f\|_\infty, \|\nabla f_0\|_\infty \le M.$$

Hence, for every $(x, y)$,

$$\begin{aligned}
|g_f(x,y)| &\le |\ell_{\mathrm{CE}}(y,f(x))| + |\ell_{\mathrm{CE}}(y,f_0(x))| + \lambda\left(\|\nabla f(x)\|_2^2 + \|\nabla f_0(x)\|_2^2\right)\\
&\le 2(\log(1+e^M)+M) + 2\lambda M^2\\
&\le 2(\log(1+e^M)+M) + 2M^2 =: M_\infty.
\end{aligned}$$

Therefore

$$\|\widetilde{g}_f\|_\infty \le \frac{2M_\infty}{r} =: \beta_{\mathrm{Tal}}. \tag{37}$$

We next bound the second moment. From the Lipschitz estimate in Step 1,

$$|g_f(x,y)| \le |f(x)-f_0(x)| + 2M\lambda\|\nabla f(x) - \nabla f_0(x)\|_2.$$

Thus

$$\mathbb{E}[g_f^2] \le C_M\left(\|f-f_0\|_{L^2(\mu)}^2 + \|\lambda\nabla(f-f_0)\|_{L^2(\mu)}^2\right).$$

Using the strong-convexity bounds from Step 1,

$$\mathbb{E}[g_f^2] \le C_M(E(f)+E(f_0)).$$

Consequently,

$$\mathbb{E}[\widetilde{g}_f^2] = \frac{\mathbb{E}[(g_f-\mathbb{E}[g_f])^2]}{(E(f)+E(f_0)+r)^2} \le \frac{\mathbb{E}[g_f^2]}{(E(f)+E(f_0)+r)^2} \le \frac{C_M}{r} =: \sigma_{\mathrm{Tal}}^2.$$

Moreover,

$$\mathbb{E}[\widetilde{g}_f] = 0.$$

Step 4: Talagrand concentration and choosing the radius. By Talagrand's inequality, with probability at least $1-e^{-t}$,

$$\begin{aligned}
\sup_{\widetilde{g}\in\widetilde{\mathcal{T}}_r^{\mathrm{sh}}} \frac{1}{2n}\sum_{i=1}^{2n}\widetilde{g}(X_i,Y_i) &\le 2\sup_{\widetilde{g}\in\widetilde{\mathcal{T}}_r^{\mathrm{sh}}}\mathbb{E}\left[\frac{1}{2n}\sum_{i=1}^{2n}\widetilde{g}(X_i',Y_i')\right] + \sqrt{\frac{2t\sigma_{\mathrm{Tal}}^2}{n}} + \frac{2t\beta_{\mathrm{Tal}}}{n}\\
&\le \frac{16\phi(r)}{r} + C_M\sqrt{\frac{t}{nr}} + \frac{C_M t}{nr} =: \psi(r).
\end{aligned}$$

Choose

$$r_0 := C_M' \max\left\{r^\star, \frac{t}{n}\right\},$$

where $C_M'$ is large enough. Since $\phi$ is sub-root and $r^\star$ is its critical radius, the first term satisfies

$$\frac{16\phi(r_0)}{r_0} \le \frac{1}{8}.$$

The choice $r_0 \gtrsim t/n$ also gives

$$C_M\sqrt{\frac{t}{nr_0}} \le \frac{1}{6}, \qquad \frac{C_M t}{nr_0} \le \frac{1}{6},$$

after increasing $C_M'$ if necessary. Therefore,

$$\psi(r_0) < \frac{1}{2}.$$

Step 5: Concluding the shifted oracle bound. Pick $r = r_0$. On the event above, for every $f \in \mathcal{F}$,

$$\frac{\mathbb{G}_n(f)}{E(f) + E(f_0) + r_0} = \frac{1}{2n} \sum_{i=1}^{2n} \widetilde{g}_f(X_i, Y_i) \le \frac{1}{2},$$

where

$$\mathbb{G}_n(f) := \Big[J_\lambda(f) - \widehat{J}_{\lambda,\mathcal{D}}(f)\Big] - \Big[J_\lambda(f_0) - \widehat{J}_{\lambda,\mathcal{D}}(f_0)\Big].$$

Thus

$$\mathbb{G}_n(f) \le \frac{1}{2}E(f) + \frac{1}{2}E(f_0) + \frac{1}{2}r_0.$$

Now take $f = \widehat{f}_{\lambda,\mathcal{F}}$. By empirical optimality,

$$\widehat{J}_{\lambda,\mathcal{D}}(\widehat{f}_{\lambda,\mathcal{F}}) \le \widehat{J}_{\lambda,\mathcal{D}}(f_0).$$

Therefore

$$\begin{aligned}
E(\widehat{f}_{\lambda,\mathcal{F}}) &= J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - J_\lambda(f_\lambda) \\
&\le \Big[J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - \widehat{J}_{\lambda,\mathcal{D}}(\widehat{f}_{\lambda,\mathcal{F}})\Big] - \Big[J_\lambda(f_0) - \widehat{J}_{\lambda,\mathcal{D}}(f_0)\Big] + E(f_0) \\
&= \mathbb{G}_n(\widehat{f}_{\lambda,\mathcal{F}}) + E(f_0) \\
&\le \frac{1}{2}E(\widehat{f}_{\lambda,\mathcal{F}}) + \frac{3}{2}E(f_0) + \frac{1}{2}r_0.
\end{aligned}$$

Rearranging gives

$$E(\widehat{f}_{\lambda,\mathcal{F}}) \le 3E(f_0) + r_0.$$

Since $r_0 \lesssim \max\{r^\star, t/n\}$, we conclude that

$$J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - J_\lambda(f_\lambda) \lesssim E(f_0) + r^\star + \frac{t}{n}.$$

This proves the theorem. $\square$

### A.4. Proof of Theorem 3.1

*Proof.* Let

$$\mathcal{F} := \mathcal{F}_M(L_n, W_n, S_n, B_n).$$

By Proposition A.4, there exists

$$f_0 = f_{\mathcal{F}}^{\mathrm{app}} \in \mathcal{F}$$

such that

$$\|f_0 - f^\star\|_{H^1(\mu)}^2 \le C_{\mathrm{app}} N^{-\frac{2(s-1)}{d}}. \tag{38}$$

Define

$$\delta_N := N^{-\frac{s-1}{d}}.$$

Then

$$\|f_0 - f^\star\|_{L^2(\mu)} \le C_{\mathrm{app}}^{1/2}\delta_N, \qquad \|\nabla(f_0 - f^\star)\|_{L^2(\mu)} \le C_{\mathrm{app}}^{1/2}\delta_N.$$

We first bound the comparator excess

$$E(f_0) := J_\lambda(f_0) - J_\lambda(f_\lambda).$$

Since $f_\lambda$ minimizes $J_\lambda$, $E(f_0) \ge 0$. Moreover,

$$E(f_0) \le J_\lambda(f_0) - J_\lambda(f^\star) + J_\lambda(f^\star) - J_\lambda(f_\lambda).$$

Step 1: Bound $J_\lambda(f_0) - J_\lambda(f^\star)$. Since $f^\star$ minimizes the population cross-entropy risk and

$$\partial_u^2 \ell_{\mathrm{CE}}(y, u) = \sigma(u)(1 - \sigma(u)) \le \frac{1}{4},$$

we have

$$L_{\mathrm{CE}}(f_0) - L_{\mathrm{CE}}(f^\star) \le \frac{1}{8}\|f_0 - f^\star\|^2_{L^2(\mu)} \le \frac{1}{8}C_{\mathrm{app}}\delta_N^2.$$

For the Sobolev penalty, using

$$\|\nabla f_0\|_{L^\infty(\mu)} \le M, \qquad \|\nabla f^\star\|_{L^\infty(\mu)} \le M,$$

we obtain

$$\begin{aligned}
\lambda \left| \|\nabla f_0\|^2_{L^2(\mu)} - \|\nabla f^\star\|^2_{L^2(\mu)} \right| &\le \lambda \int \left(\|\nabla f_0\|_2 + \|\nabla f^\star\|_2\right) \|\nabla (f_0 - f^\star)\|_2 \, d\mu \\
&\le 2M\lambda \|\nabla (f_0 - f^\star)\|_{L^2(\mu)} \\
&\le 2M C_{\mathrm{app}}^{1/2} \lambda \delta_N.
\end{aligned}$$

Therefore

$$J_\lambda(f_0) - J_\lambda(f^\star) \le \frac{1}{8}C_{\mathrm{app}}\delta_N^2 + 2MC_{\mathrm{app}}^{1/2}\lambda\delta_N. \tag{39}$$

<u>Step 2: Bound $J_\lambda(f^\star) - J_\lambda(f_\lambda)$.</u> Since $f^\star$ minimizes $L_{\mathrm{CE}}$,

$$L_{\mathrm{CE}}(f^\star) - L_{\mathrm{CE}}(f_\lambda) \le 0.$$

Thus

$$J_\lambda(f^\star) - J_\lambda(f_\lambda) \le \lambda \left( \|\nabla f^\star\|^2_{L^2(\mu)} - \|\nabla f_\lambda\|^2_{L^2(\mu)} \right).$$

Let

$$\delta_\lambda := f_\lambda - f^\star.$$

Then

$$\|\nabla f^\star\|^2_{L^2(\mu)} - \|\nabla f_\lambda\|^2_{L^2(\mu)} = -2\langle \nabla f^\star, \nabla \delta_\lambda \rangle_{L^2(\mu)} - \|\nabla \delta_\lambda\|^2_{L^2(\mu)}.$$

Hence

$$J_\lambda(f^\star) - J_\lambda(f_\lambda) \le 2\lambda \left| \langle \nabla f^\star, \nabla \delta_\lambda \rangle_{L^2(\mu)} \right|.$$

Using the same Green identity as in Lemma 3.4,

$$\left| \langle \nabla f^\star, \nabla \delta_\lambda \rangle_{L^2(\mu)} \right| = \left| \langle R_0^\star, \delta_\lambda \rangle_{L^2(\mu)} \right| \le \|R_0^\star\|_{L^2(\mu)} \|\delta_\lambda\|_{L^2(\mu)}.$$

Let

$$B_0 := \|R_0^\star\|_{L^2(\mu)}.$$

By Lemma 3.4,

$$\|\delta_\lambda\|_{L^2(\mu)} \le \sqrt{\beta}\lambda.$$

Therefore

$$J_\lambda(f^\star) - J_\lambda(f_\lambda) \le 2B_0\sqrt{\beta}\,\lambda^2. \tag{40}$$

Combining (39) and (40), we obtain

$$E(f_0) \le \frac{1}{8}C_{\mathrm{app}}\delta_N^2 + 2MC_{\mathrm{app}}^{1/2}\lambda\delta_N + 2B_0\sqrt{\beta}\lambda^2. \tag{41}$$

<u>Step 3: Apply the shifted oracle inequality.</u> Apply Theorem 3.5 with $t = 2\log n$. With probability at least $1 - n^{-2}$,

$$J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - J_\lambda(f_\lambda) \le C_{\mathrm{or}} \left( E(f_0) + r^\star + \frac{\log n}{n} \right).$$

Using (41), we get

$$J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - J_\lambda(f_\lambda) \le C_{\mathrm{or}} \left[ \frac{1}{8}C_{\mathrm{app}}\delta_N^2 + 2MC_{\mathrm{app}}^{1/2}\lambda\delta_N + 2B_0\sqrt{\beta}\lambda^2 + r^\star + \frac{\log n}{n} \right]. \tag{42}$$

Step 4: Convert excess risk to $L^2$ and gradient errors. By Lemma 3.3, for any $f, g \in \mathcal{H}_M$,

$$J_\lambda(f) - J_\lambda(g) - DJ_\lambda(g)[f-g] \geq \frac{c_{\min}}{2}\|f-g\|^2_{L^2(\mu)} + \lambda\|\nabla(f-g)\|^2_{L^2(\mu)}.$$

Since $f_\lambda$ is the population minimizer, the first-order optimality condition gives

$$DJ_\lambda(f_\lambda)[\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda] \geq 0.$$

Therefore

$$\frac{c_{\min}}{2}\|\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda\|^2_{L^2(\mu)} + \lambda\|\nabla(\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda)\|^2_{L^2(\mu)} \leq J_\lambda(\widehat{f}_{\lambda,\mathcal{F}}) - J_\lambda(f_\lambda).$$

Combining this with (42), we obtain

$$\|\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda\|^2_{L^2(\mu)} \leq \frac{2C_{\mathrm{or}}}{c_{\min}}\left[\frac{1}{8}C_{\mathrm{app}}\delta_N^2 + 2MC_{\mathrm{app}}^{1/2}\lambda\delta_N + 2B_0\sqrt{\beta}\lambda^2 + r^\star + \frac{\log n}{n}\right], \tag{43}$$

and

$$\|\nabla(\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda)\|^2_{L^2(\mu)} \leq \frac{C_{\mathrm{or}}}{\lambda}\left[\frac{1}{8}C_{\mathrm{app}}\delta_N^2 + 2MC_{\mathrm{app}}^{1/2}\lambda\delta_N + 2B_0\sqrt{\beta}\lambda^2 + r^\star + \frac{\log n}{n}\right]. \tag{44}$$

Step 5: Add the regularization bias. By Lemma 3.4,

$$\|f_\lambda - f^\star\|^2_{L^2(\mu)} \leq \beta\lambda^2, \qquad \|\nabla(f_\lambda - f^\star)\|^2_{L^2(\mu)} \leq \beta\lambda.$$

Using the triangle inequality,

$$\|\widehat{f}_{\lambda,\mathcal{F}} - f^\star\|^2_{L^2(\mu)} \leq 2\|\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda\|^2_{L^2(\mu)} + 2\|f_\lambda - f^\star\|^2_{L^2(\mu)},$$

and

$$\|\nabla(\widehat{f}_{\lambda,\mathcal{F}} - f^\star)\|^2_{L^2(\mu)} \leq 2\|\nabla(\widehat{f}_{\lambda,\mathcal{F}} - f_\lambda)\|^2_{L^2(\mu)} + 2\|\nabla(f_\lambda - f^\star)\|^2_{L^2(\mu)}.$$

Therefore

$$\begin{aligned}\|\widehat{f}_{\lambda,\mathcal{F}} - f^\star\|^2_{H^1(\mu)} \leq{}& \left(\frac{4C_{\mathrm{or}}}{c_{\min}} + \frac{2C_{\mathrm{or}}}{\lambda}\right)\left[\frac{1}{8}C_{\mathrm{app}}\delta_N^2 + 2MC_{\mathrm{app}}^{1/2}\lambda\delta_N + 2B_0\sqrt{\beta}\lambda^2 + r^\star + \frac{\log n}{n}\right] \\ &+ 2\beta\lambda^2 + 2\beta\lambda.\end{aligned} \tag{45}$$

Step 6: Choose $\lambda$ and $N$. The critical radius satisfies

$$r^\star \leq C_{\mathrm{rad}}\frac{N(\log N + \log n)}{n},$$

where $C_{\mathrm{rad}}$ depends only on the fixed sieve envelope $M$ and fixed depth $L = O(1)$.

Choose

$$\lambda \asymp \delta_N = N^{-\frac{s-1}{d}}.$$

Since $c_{\min}$, $M$, $B_0$, $\beta$, $C_{\mathrm{app}}$, and $C_{\mathrm{or}}$ are independent of $n$, the leading terms in (45) are bounded by

$$C\left[\delta_N + \frac{r^\star}{\delta_N} + \frac{\log n}{n\delta_N}\right].$$

Since $r^\star \gtrsim N/n$ up to logarithmic factors, the term

$$\frac{\log n}{n\delta_N}$$

is dominated by $r^\star/\delta_N$. Hence

$$\|\widehat{f}_{\lambda,\mathcal{F}} - f^\star\|^2_{H^1(\mu)} \leq C\left[N^{-\frac{s-1}{d}} + \frac{N^{1+\frac{s-1}{d}}(\log N + \log n)}{n}\right].$$

Balancing the two terms gives

$$N^{-\frac{s-1}{d}} \asymp \frac{N^{1+\frac{s-1}{d}}}{n},$$

and therefore

$$N \asymp n^{\frac{d}{d+2s-2}}.$$

Consequently,

$$\lambda \asymp N^{-\frac{s-1}{d}} \asymp n^{-\frac{s-1}{d+2s-2}}.$$

Substituting this choice yields

$$\|\widehat{f}_{\lambda,\mathcal{F}} - f^\star\|^2_{H^1(\mu)} \leq Cn^{-\frac{s-1}{d+2s-2}} \log n.$$

This proves the theorem. □

## B. Proof of Theorem 4.1: minimax lower bound

*Proof.* Step 1: Local packing on $\mathcal{P}$ with enough separation. Let $\mathcal{V}$ be a finite index set. We construct a finite subset $\{p_v, q_v\}_{v\in\mathcal{V}}$ of $\mathcal{P}$ on which the corresponding log-density ratio $f_v$ has a non-trivial $H^1$-separation. We achieve this goal by first defining some saperating functions and then correct them to be the log-density ratio.

Let $\eta : \mathbb{R} \to \mathbb{R}$ be the one-dimensional $C^\infty$ bump

$$\eta(t) := \begin{cases} \exp\big(-\frac{1}{t(1-t)}\big), & t \in (0,1), \\ 0, & t \notin (0,1), \end{cases}$$

and define

$$\varphi(x) := \prod_{i=1}^{d} \eta(x_i), \qquad x = (x_1, \ldots, x_d) \in \mathbb{R}^d.$$

Then $\varphi \in C^\infty(\mathbb{R}^d)$, $\varphi \geq 0$, $\nabla\varphi \not\equiv 0$ , and $\mathrm{supp}(\varphi) \subset [0,1]^d$.

For an integer $m \geq 1$ we partition $[0,1]^d$ into $m^d$ disjoint cubes of side length $\asymp 1/m$ and choose points $\{x_j\}_{j\in[m]^d} \subset \mathbb{R}^d$ such that the cubes $x_j + [0, 1/m]^d$ are disjoint and contained in $[0,1]^d$. For each $j \in [m]^d$ define the rescaled bump by

$$\varphi_{m,j}(x) := \varphi\big(m(x - x_j)\big).$$

Using the change of variables $u = m(x - x_j)$ and standard Sobolev scaling, we obtain

$$\|\varphi_{m,j}\|^2_{L^2(\mu_0)} \asymp m^{-d}, \qquad \|\nabla\varphi_{m,j}\|^2_{L^2(\mu_0)} \asymp m^{2-d}, \qquad \|\varphi_{m,j}\|^2_{H^s(\mu_0)} \asymp m^{2s-d}, \tag{46}$$

where the implicit constants depend only on $\varphi$ and the density bounds for $\mu_{p,q}$.

By the Varshamov-Gilbert bound(Takezawa, 2005), there exists a subset

$$\big\{\tau^{(1)}, \ldots, \tau^{(N)}\big\} \subset \{0,1\}^{m^d}$$

such that

$$N \geq 2^{m^d/8}, \qquad \|\tau^{(v)} - \tau^{(v')}\|^2_2 \geq \frac{m^d}{8} \quad \text{for all } v \neq v'.$$

Fix an amplitude $a_m > 0$ (to be specified below) and define

$$\tilde{f}_v(x) := a_m \sum_{j=1}^{m^d} \tau_j^{(v)}\, \varphi_{m,j}(x), \qquad v = 1, \ldots, N.$$

For each such $\tilde{f}_v$ we define an associated pair of probability densities $(p_v, q_v)$ on $\Omega$ as follows. First set

$$\tilde{p}_v(x) := \sigma\big(-\tilde{f}_v(x)\big), \qquad \tilde{q}_v(x) := \sigma\big(\tilde{f}_v(x)\big), \qquad x \in \Omega,$$

and then normalize:

$$p_v(x) := \frac{\sigma\big(-\tilde{f}_v(x)\big)}{Z_p}, \qquad q_v(x) := \frac{\sigma\big(\tilde{f}_v(x)\big)}{Z_q} \qquad Z_p = \int_\Omega \sigma\big(-\tilde{f}_v(x)\big)\,\mathrm{d}u, Z_q = \int_\Omega \sigma\big(\tilde{f}_v(x)\big)\,\mathrm{d}u$$

By construction, $p_v$ and $q_v$ are probability densities on $\Omega$. We have $f_v = \log\frac{q_v}{p_v} = \tilde{f}_v - \log Z_p - \log Z_q$. We need to prove that they satisfy Assumption 2.1.

Observe that $|\log Z_p|, |\log Z_q| \le \sup_{x\in\Omega} |\tilde{f}_v(x)|$. Since the supports of the $\varphi_{m,j}$ are disjoint, we have

$$\sup_{x\in\Omega} |f_v(x)| \le 3 \sup_{x\in\mathbb{R}^d} |\tilde{f}_v(x)| \le 3a_m \max_j \sup_{x\in\mathbb{R}^d} |\varphi_{m,j}(x)| \lesssim a_m, \tag{47}$$

$$\|f_v\|^2_{H^s(\mu_0)} \lesssim a_m^2 \sum_{j=1}^{m^d} \|\varphi_{m,j}\|^2_{H^s(\mu_{p_k,q_k})} + 2\sup_{x\in\mathbb{R}^d} |\tilde{f}_k(x)| \lesssim a_m^2\, m^d\, m^{2s-d} + 2a_m^2 = a_m^2 m^{2s} + a_m^2. \tag{48}$$

Hence by choosing $a_m$ of order $m^{-s}$ with a sufficiently small constant, we can ensure that

$$\|f_v\|_{H^s(\mu_0)} < \infty, \qquad \sup_{x\in\Omega} |f_v(x)| \le M.$$

Since $f_v$ is bounded and smooth with compact support, it satisfies the condition 3 in Assumption 2.1. Also $(p_v, q_v)$ are strictly positive and $C^2$. Moreover, since the support of $\varphi$ has distance $d \simeq 1/m$ away from $\partial[0,1]^d$, $p_v$ and $q_v$ is constant on $\partial\Omega$ and thus satisfy Neumann boundary condition. Since $\frac{1}{2}p_v + \frac{1}{2}q_v$ is always the uniform distribution, following the binary classification form (6), $f_v$ induces the joint distribution $P_v$ for $(X, Y)$ by

$$X \sim \mu_0, \qquad Y \mid X = x \sim \mathrm{Bernoulli}(\sigma(f_v(x))).$$

Since we verify all the conditions, we have $(p_v, q_v) \in \mathcal{P}$.

Finally we are ready to show the pairwise separation results in $H^1$. For $v \ne v'$, using disjoint supports and (46),

$$\begin{aligned}
\|\nabla f_v - \nabla f_{v'}\|^2_{L^2(\mu_0)} &= a_m^2 \sum_{j=1}^{m^d} \big(\tau_j^{(k)} - \tau_j^{(v')}\big)^2 \|\nabla\varphi_{m,j}\|^2_{L^2(\mu)} \\
&\gtrsim a_m^2 \Big(\sum_{j=1}^{m^d} \big|\tau_j^{(k)} - \tau_j^{(v')}\big|^2\Big) m^{2-d} \\
&\gtrsim a_m^2\,(m^d)\, m^{2-d} \asymp a_m^2 m^2 \asymp m^{-2s} m^2 = m^{-2(s-1)}.
\end{aligned}$$

So we deduce that, there exists constant $c_1 > 0$ so that

$$\|f_v - f_{v'}\|^2_{H^1(\mu_0)} = \|f_v - f_{v'}\|^2_{L^2(\mu_0)} + \|\nabla f_v - \nabla f_{v'}\|^2_{L^2(\mu_0)} \ge c_1\, m^{-2(s-1)}, \qquad v \ne v'. \tag{49}$$

Hence the family $\{f_1, \dots, f_N\}$ is separated in $H^1(\mu_0)$ with separation order of $m^{-2(s-1)}$.

Step 2: KL upper bound for the joint measure $P_v$. Consider two candidates $P_v$ and $P_{v'}$ on $(X, Y)$ pair. Conditioned on $x$, the conditional KL divergence is

$$\begin{aligned}
\mathrm{KL}_x(f, g) &:= \mathrm{KL}\Big(\mathrm{Bern}(\sigma(f(x)))\Big\|\mathrm{Bern}(\sigma(g(x)))\Big) \\
&= \mathbb{E}_{Y\sim\mathrm{Bern}(\sigma(f(x)))}\big[\ell_{\mathrm{CE}}(f(x); Y) - \ell_{\mathrm{CE}}(g(x); Y)\big] \\
&= G_x(f(x)) - G_x(g(x)).
\end{aligned} \tag{50}$$

Note that for every $x$ and every $t \in [0,1]$,

$$c_{\min} \;\le\; G_x''\big(f(x) + t(g(x) - f(x))\big) \;\le\; c_{\max}.$$

So by Taylor's theorem with integral remainder, for each fixed $x$ we have

$$\frac{c_{\min}}{2}\,(f(x) - g(x))^2 \;\le\; G_x(f(x)) - G_x(g(x)) \;\le\; \frac{c_{\max}}{2}\,(f(x) - g(x))^2.$$

Since $P_v$ and $P_{v'}$ share the same marginal distribution on $x$ part, taking expectation for $x$ with respect to $\mu_0$, we have

$$c_{\min}\|f - g\|^2_{L^2(\mu_0)} \lesssim \mathrm{KL}(P_v \| P_{v'}) \lesssim c_{\max}\|f - g\|^2_{L^2(\mu_0)}. \tag{51}$$

Now consider $2n$ pairs $(X_i, Y_i)_{i=1}^{2n}$ drawn from i.i.d. $P_v$. Denote the product measure as $P_v^{2n}$. For each $v \neq v'$,

$$\mathrm{KL}(P_v^{2n} \| P_{v'}^{2n}) = 2n\,\mathrm{KL}(P_v \| P_{v'}) \lesssim 2n\,\|f_v - f_{v'}\|^2_{L^2(\mu_0)} \tag{52}$$

Note that $\sum_{j=1}^{m^d} \big|\tau_j^{(k)} - \tau_j^{(v')}\big|^2 \le m^d$, so we have

$$\begin{aligned}
\|f_v - f_{v'}\|^2_{L^2(\mu_0)} &\le \|\tilde f_k - \tilde f_{v'}\|^2_{L^2(\mu_0)} + |\log Z_k - \log Z_{v'}|^2 \\
&\le a_m^2 \sum_{j=1}^{m^d} \big(\tau_j^{(k)} - \tau_j^{(v')}\big)^2 \|\varphi_{m,j}\|^2_{L^2(\mu_0)} + 2\sup_{x \in \mathbb{R}^d} |\tilde f_k(x)|^2 \\
&\lesssim a_m^2 \Big(\sum_{j=1}^{m^d} \big|\tau_j^{(k)} - \tau_j^{(v')}\big|^2\Big) m^{-d} + a_m^2 \\
&\lesssim a_m^2\,(m^d)\,m^{-d} + a_m^2 \asymp a_m^2 \asymp m^{-2s}.
\end{aligned} \tag{53}$$

Combining (52) with (53) we get

$$\mathrm{KL}(P_v^{2n} \| P_{v'}^{2n}) \lesssim n\,m^{-2s}. \tag{54}$$

<u>Step 3: Local Fano reduction and minimax lower bound.</u> Recall the finite family $\{f_1, \ldots, f_N\}$ constructed in Step 1.

Let $V$ be a random index uniformly distributed on $\{1, \ldots, N\}$, and conditional on $V = v$ let $(X_i, Y_i)\}_{i=1}^{2n} \sim P_v$. Any estimator $\widehat f = \psi(\{(X_i, Y_i)\}_{i=1}^{2n})$ induces a testing rule

$$\widehat V(\{(X_i, Y_i)\}_{i=1}^{2n}) := \operatorname*{arg\,min}_{0 \le v \le N} \big\|\psi(\{(X_i, Y_i)\}_{i=1}^{2n}) - f_v\big\|_{H^1(\mu_0)},$$

with error probability $\mathbb{P}(\widehat V \neq V)$. By the $H^1$-separation in (49), we have

$$\|f_v - f_{v'}\|^2_{H^1(\mu_0)} \;\ge\; c_1\,m^{-2(s-1)}, \qquad v \neq v'.$$

if $\widehat V \neq V$, then $\big\|\widehat f - f_V\big\|_{H^1(\mu_0)} \ge \big\|\widehat f - f_{\widehat V}\big\|_{H^1(\mu_0)}$, so we have $\big\|\widehat f - f_V\big\|_{H^1(\mu_0)} \ge \frac{c_1}{2} m^{-2(s-1)}$. Hence

$$\sup_{0 \le v \le N} \mathbb{E}_{P_k}\big[\|\widehat f - f_v\|^2_{H^1(\mu_0)}\big] \;\ge\; \mathbb{E}\big[\|\widehat f - f_V\|^2_{H^1(\mu_0)}\big] \;\ge\; \frac{c_1}{2}\,m^{-2(s-1)}\mathbb{P}(\widehat V \neq V). \tag{55}$$

The remaining step is to bound $\mathbb{P}(\widehat V \neq V)$. From Step 2, Applying KL bound (54) yields

$$\mathrm{KL}\big(P_v \big\| P_{v'}\big) = \mathrm{KL}\big(P_v^{2n} \big\| P_{v'}^{2n}\big) \;\lesssim\; 2n\,\|f_v - f_{v'}\|^2_{L^2(\mu_0)} \;\lesssim\; n\,m^{-2s}.$$

So if we choose

$$m \;\gtrsim\; n^{1/(2s+d)},$$

we will have $\mathrm{KL}\big(P_v \big\| P_{v'}\big) \lesssim m^d$. Therefore, by local Fano inequality (Takezawa, 2005) and $\log N \asymp m^d$, we will have

$$\begin{aligned}\mathbb{P}(\widehat{V} \neq V) &\geq 1 - \frac{I(V; \{(X_i, Y_i)\}_{i=1}^{2n}) + \log 2}{\log N} \\ &\geq 1 - \frac{1/N^2 \times \sum_{v,v'} \mathrm{KL}(P_v \| P_{v'}) + \log 2}{\log N} \geq \frac{1}{2},\end{aligned} \tag{56}$$

where the last inequality holds if we take $a_m$ to be sufficiently small. Combining (55) and (56), and taking $\inf$ for any estimator $\psi$ , we have

$$\inf_{\psi} \sup_{0 \leq v \leq N} \mathbb{E}_{P_v}\left[\|\psi(\{(X_i, Y_i)\}_{i=1}^{2n}) - f_v\|_{H^1(\mu_0)}^2\right] \gtrsim n^{-\frac{2(s-1)}{2s+d}}.$$

$\square$

# C. Proof for the extension to diffusion models

In this section we prove Theorem 5.2. We work on the bounded time-space cylinder

$$C_{2R} := [t_0, T] \times B_{2R}, \qquad B_{2R} := \{x \in \mathbb{R}^d : \|x\| \leq 2R\},$$

and use a projection-rescaling argument so that the neural network is trained on the fixed compact spatial domain $B_1$.

## C.1. Notation

Throughout Appendix C, for each truncation radius $R \geq 1$, we use the $R$-dependent clipped-gradient time-space class

$$\mathcal{F}_R := \mathcal{F}_{M_R}^{(d+1)}(L, W, S, B), \qquad D = d + 1, \qquad M_R := C_M(1 + R),$$

on the fixed input domain

$$C_1 := [t_0, T] \times B_1 \subset \mathbb{R}^{d+1}.$$

The input variable is $(t, \bar{x})$, and time is treated as an ordinary network input. By definition of $\mathcal{F}_R$,

$$\|f\|_{L^\infty(C_1)} \leq M_R, \qquad \|\nabla_{\bar{x}} f\|_{L^\infty(C_1)} \leq M_R, \qquad f \in \mathcal{F}_R.$$

**Time-dependent label-augmented model and truncated population risk.** For each $t \in [t_0, T]$, define a joint distribution $\mu_t^{x,y}$ on $\mathbb{R}^d \times \{0, 1\}$ by

$$Y \sim \text{Bernoulli}(1/2), \qquad X \mid (Y = 1) \sim q_t, \qquad X \mid (Y = 0) \sim p_t. \tag{57}$$

The time-$t$ mixture marginal of $X$ is

$$\mu_t(dx) = \rho_t(x)\, dx, \qquad \rho_t(x) := \frac{1}{2}(p_t(x) + q_t(x)).$$

We also write

$$\eta_t(x) := \mathbb{P}(Y = 1 \mid X = x) = \frac{q_t(x)}{p_t(x) + q_t(x)}.$$

Recall the pointwise $H^1$-loss density in (15). For any function $h : [t_0, T] \times \mathbb{R}^d \to \mathbb{R}$, define the truncated population functional on $C_{2R}$ by

$$J_{\lambda,2R}(h) := \int_{t_0}^{T} \mathbb{E}_{(X,Y) \sim \mu_t^{x,y}}\Big[\mathbf{1}\{X \in B_{2R}\}\Big(\ell_{\mathrm{CE}}(Y, h(t, X)) + \lambda \|\nabla_x h(t, X)\|^2\Big)\Big]\, dt. \tag{58}$$

1058 **Projection-rescaling and the induced estimator on $B_{2R}$.** Recall the projection

$$\Pi_{2R}(x) = \frac{x}{\max\{1, \|x\|/(2R)\}} \in B_{2R}, \qquad \bar{x} = \Pi_{2R}(x)/(2R) \in B_1.$$

1059 Given any $f : [t_0, T] \times B_1 \to \mathbb{R}$, define its pull-back to $C_{2R}$ by

$$(\mathsf{P}_R f)(t, x) := f\Big(t, \frac{x}{2R}\Big), \qquad x \in B_{2R}. \tag{59}$$

1060 A direct differentiation gives the gradient relation on $B_{2R}$:

$$\nabla_x (\mathsf{P}_R f)(t, x) = \frac{1}{2R}\, \nabla_{\bar{x}} f\Big(t, \frac{x}{2R}\Big). \tag{60}$$

1061 Define the rescaled target on $C_1$ by

$$f_R^\star(t, \bar{x}) := f^\star(t, 2R\bar{x}), \qquad (t, \bar{x}) \in C_1. \tag{61}$$

1062 Then for $x \in B_{2R}$,

$$f^\star(t, x) = f_R^\star\Big(t, \frac{x}{2R}\Big), \qquad \nabla_x f^\star(t, x) = \frac{1}{2R} \nabla_{\bar{x}} f_R^\star\Big(t, \frac{x}{2R}\Big). \tag{62}$$

1063 The associated pulled-back network class on $C_{2R}$ is

$$\mathcal{F}_{2R} := \{\mathsf{P}_R f : f \in \mathcal{F}_R\}. \tag{63}$$

1064 For every $h = \mathsf{P}_R f \in \mathcal{F}_{2R}$, the definition of $\mathcal{F}_R$ and (60) imply

$$\|h\|_{L^\infty(C_{2R})} \le M_R, \qquad \|\nabla_x h\|_{L^\infty(C_{2R})} \le \frac{M_R}{2R} \le C_M, \qquad R \ge 1. \tag{64}$$

1065 **Norms.** On $C_{2R}$, we use the $\mu_t$-weighted spatial Sobolev norm

$$\|g\|^2_{H^{1,x}(C_{2R})} := \int_{t_0}^T \int_{B_{2R}} \Big( |g(t, x)|^2 + \|\nabla_x g(t, x)\|^2 \Big)\, d\mu_t(x)\, dt. \tag{65}$$

1066 On the fixed unit cylinder $C_1$, we use the Lebesgue spatial Sobolev norm

$$\|u\|^2_{H^{1,\bar{x}}(C_1;\mathrm{Leb})} := \int_{t_0}^T \int_{B_1} \Big( |u(t, \bar{x})|^2 + \|\nabla_{\bar{x}} u(t, \bar{x})\|^2 \Big)\, d\bar{x}\, dt. \tag{66}$$

1067 For integer $s \ge 1$, we use the full time-space Sobolev norm

$$\|u\|^2_{H^s_{t,\bar{x}}(C_1;\mathrm{Leb})} := \sum_{a+|\gamma| \le s} \int_{t_0}^T \int_{B_1} \big|\partial_t^a D_{\bar{x}}^\gamma u(t, \bar{x})\big|^2\, d\bar{x}\, dt. \tag{67}$$

1068 Here $H^{1,\bar{x}}$ measures only spatial derivatives, while $H^s_{t,\bar{x}}$ measures derivatives in both time and space.

1069 For a function $g : C_{2R} \to \mathbb{R}$, define its spatially rescaled version by

$$g^{(R)}(t, \bar{x}) := g(t, 2R\bar{x}), \qquad (t, \bar{x}) \in C_1. \tag{68}$$

1070 In particular, for the target $f^\star$, define

$$\mathcal{A}_s(R) := (2R)^d \big\|(f^\star)^{(R)}\big\|^2_{H^s_{t,\bar{x}}(C_1;\mathrm{Leb})}. \tag{69}$$

1071 **Bounded Sobolev classes on truncated cylinders.** For $R > 0$ and $M > 0$, define the bounded Sobolev class on $C_{2R}$ by

$$\mathcal{H}_M(C_{2R}) := \Big\{ h : C_{2R} \to \mathbb{R} \;\Big|\; h \in H^{1,x}(C_{2R}),\ |h(t, x)| \le M \text{ for a.e. } (t, x) \in C_{2R} \Big\}. \tag{70}$$

### C.2. Uniform bounds on the pulled-back class and auxiliary lemmas

We first collect the Gaussian-convolution bounds.

**Lemma C.1** (Basic consequences of the regular VP assumption)**.** *Suppose Assumption 5.1 holds. Define*

$$\nu_{p,t} := (\alpha_t)_{\#}p_0, \qquad \nu_{q,t} := (\alpha_t)_{\#}q_0, \qquad R_\nu := C_{\mathrm{VP}}R_{\mathrm{init}}.$$

*Then*

$$\mathrm{supp}(\nu_{p,t}) \cup \mathrm{supp}(\nu_{q,t}) \subseteq B_{R_\nu}, \qquad \underline{\sigma} \le \sigma_t \le \overline{\sigma}, \qquad t \in [t_0, T].$$

*Moreover, there exist constants $C, c, K < \infty$, depending only on*

$$d, s, t_0, T, R_{\mathrm{init}}, \underline{\sigma}, \overline{\sigma}, C_{\mathrm{VP}},$$

*such that the following bounds hold uniformly over $t \in [t_0, T]$.*

*First, the mixture measure has sub-Gaussian tails: for every $u \ge 0$,*

$$\mu_t(\{x : \|x\| > u\}) \le C \exp\{-c(u - R_\nu)_+^2\}. \tag{71}$$

*More generally, for every $m \ge 0$,*

$$\int_{\{\|x\|>u\}} (1 + \|x\|)^m \, d\mu_t(x) \le C_m (1 + u)^m \exp\{-c_m (u - R_\nu)_+^2\}. \tag{72}$$

*Second, the log-density ratio has at most linear growth:*

$$|f^\star(t, x)| \le C(1 + \|x\|). \tag{73}$$

*Its spatial gradient is uniformly bounded:*

$$\|\nabla_x f^\star(t, x)\| \le C. \tag{74}$$

*Finally, for every pair of multi-indices $(a, b)$ satisfying $a + |b| \le s$,*

$$\left|\partial_t^a D_x^b f^\star(t, x)\right| \le C(1 + \|x\|)^K. \tag{75}$$

*Proof.* The support statement follows directly from the VP representation. Since $\mathrm{supp}(p_0) \cup \mathrm{supp}(q_0) \subseteq B_{R_{\mathrm{init}}}$ and

$$|\alpha_t| \le C_{\mathrm{VP}},$$

we have

$$\mathrm{supp}((\alpha_t)_{\#}p_0) \cup \mathrm{supp}((\alpha_t)_{\#}q_0) \subseteq B_{C_{\mathrm{VP}}R_{\mathrm{init}}} = B_{R_\nu}.$$

We first prove the tail bounds. If $Z_0$ is supported in $B_{R_{\mathrm{init}}}$, then

$$Z_t = \alpha_t Z_0 + \sigma_t \xi$$

satisfies

$$\|Z_t\| \le R_\nu + \overline{\sigma}\|\xi\|.$$

Therefore

$$\mathbb{P}(\|Z_t\| > u) \le \mathbb{P}\left(\|\xi\| > \frac{(u - R_\nu)_+}{\overline{\sigma}}\right) \le C \exp\{-c(u - R_\nu)_+^2\}.$$

This holds for both $p_t$ and $q_t$, hence for their mixture $\mu_t$. The polynomial tail-moment bound (72) follows from the same Gaussian tail estimate by integration by parts.

Next we study the density ratio. Write $\sigma = \sigma_t$. Since

$$p_t = (\alpha_t)_{\#}p_0 * \mathcal{N}(0, \sigma_t^2 I), \qquad q_t = (\alpha_t)_{\#}q_0 * \mathcal{N}(0, \sigma_t^2 I),$$

we have

$$p_t(x) = \int (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|x - \alpha_t z\|^2}{2\sigma^2}\right) dp_0(z),$$

and similarly for $q_t$. Expanding the square gives

$$p_t(x) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right) I_p(t, x),$$

where

$$I_p(t, x) := \int \exp\left(\frac{\alpha_t \langle x, z\rangle}{\sigma_t^2} - \frac{\alpha_t^2 \|z\|^2}{2\sigma_t^2}\right) dp_0(z).$$

Similarly,

$$q_t(x) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right) I_q(t, x),$$

where $I_q$ is defined with $q_0$. Hence the common Gaussian factor cancels in the log-density ratio:

$$f^\star(t, x) = \log I_q(t, x) - \log I_p(t, x).$$

Since $z \in B_{R_{\text{init}}}$, $\alpha_t$ is bounded, and $\sigma_t \geq \underline{\sigma}$, the exponent inside $I_p$ is bounded above and below by affine functions of $\|x\|$. Therefore

$$|\log I_p(t, x)| \leq C(1 + \|x\|), \qquad |\log I_q(t, x)| \leq C(1 + \|x\|),$$

which proves (73).

For the gradient, define the tilted probability measure

$$d\pi_{p,t,x}(z) := \frac{\exp\left(\frac{\alpha_t \langle x, z\rangle}{\sigma_t^2} - \frac{\alpha_t^2 \|z\|^2}{2\sigma_t^2}\right)}{I_p(t, x)} dp_0(z).$$

Then

$$\nabla_x \log I_p(t, x) = \frac{\alpha_t}{\sigma_t^2} \int z \, d\pi_{p,t,x}(z).$$

Since $\|z\| \leq R_{\text{init}}$, we obtain

$$\|\nabla_x \log I_p(t, x)\| \leq C.$$

The same bound holds for $I_q$. Therefore

$$\nabla_x f^\star(t, x) = \nabla_x \log I_q(t, x) - \nabla_x \log I_p(t, x)$$

is uniformly bounded, proving (74).

$\square$

By Lemma C.1,

$$\sup_{(t,\bar{x}) \in C_1} |f_R^\star(t, \bar{x})| \leq C(1 + R).$$

Moreover,

$$\nabla_{\bar{x}} f_R^\star(t, \bar{x}) = 2R \nabla_x f^\star(t, 2R\bar{x}),$$

and Lemma C.1 gives

$$\sup_{(t,\bar{x}) \in C_1} \|\nabla_{\bar{x}} f_R^\star(t, \bar{x})\| \leq C(1 + R).$$

Thus, by choosing $C_M$ sufficiently large, $M_R = C_M(1 + R)$ dominates both envelopes.

Consequently,

$$\mathcal{F}_{2R} \subseteq \mathcal{H}_{M_R}(C_{2R}), \qquad f^\star \in \mathcal{H}_{M_R}(C_{2R}).$$

Indeed, if $h = \mathsf{P}_R f \in \mathcal{F}_{2R}$, then
$$\|h\|_{L^\infty(C_{2R})} = \sup_{(t,x)\in C_{2R}} \left| f\left(t, \frac{x}{2R}\right)\right| \leq M_R,$$
because $f \in \mathcal{F}_R$. Moreover, the strict slack
$$M_R - \sup_{(t,x)\in C_{2R}} |f^\star(t,x)| \geq 1$$
will be used below when clipping the population minimizer.

Define
$$c_R := c_{\min}(R) := \frac{1}{4\cosh^2(M_R/2)}. \tag{76}$$
Since $M_R = C_M(1+R)$, there exists $C < \infty$ such that
$$c_R^{-1} \lesssim e^{CR}. \tag{77}$$

**Lemma C.2** (Existence, uniqueness, and clipping on $C_{2R}$)**.** *For every $\lambda > 0$, the population functional $J_{\lambda,2R}$ admits a unique minimizer over $\mathcal{H}_{M_R}(C_{2R})$. Denote it by $f_{\lambda,2R}$. Moreover,*
$$\|f_{\lambda,2R}\|_{L^\infty(C_{2R})} \leq \|f^\star\|_{L^\infty(C_{2R})} \leq M_R.$$

*Proof.* The existence and uniqueness follow from the same direct-method argument as Lemma A.1. Indeed, $\mathcal{H}_{M_R}(C_{2R})$ is convex and closed in the relevant $L_t^2 H_x^1$ topology. Since $C_{2R}$ is bounded and $M_R < \infty$, the $L^\infty$ constraint controls the $L^2$ part, while the Sobolev penalty controls the spatial-gradient part. Hence every minimizing sequence is weakly precompact. The cross-entropy term is convex and weakly lower semicontinuous, and the Sobolev penalty is weakly lower semicontinuous, so the minimum is attained. Uniqueness follows from the logistic curvature lower bound on $[-M_R, M_R]$, together with the strict convexity induced by the $L^2$ curvature term.

It remains to prove the clipping property. Let
$$M_R^\star := \|f^\star\|_{L^\infty(C_{2R})}.$$
By the choice of $C_M$ in $M_R = C_M(1+R)$ and Lemma C.1, we have $M_R^\star \leq M_R$. Define the pointwise clipping map
$$T(u) := (-M_R^\star) \vee (u \wedge M_R^\star).$$
For any $f \in \mathcal{H}_{M_R}(C_{2R})$, the clipped function $Tf$ also belongs to $\mathcal{H}_{M_R}(C_{2R})$, since
$$|Tf| \leq M_R^\star \leq M_R$$
and the chain rule for Lipschitz truncations gives
$$\|\nabla_x Tf\|_{L^2(C_{2R})} \leq \|\nabla_x f\|_{L^2(C_{2R})}.$$

Moreover, since $\eta_t = \sigma(f^\star(t,\cdot))$, the conditional logistic risk is pointwise minimized at $f^\star(t,x)$. Because $|f^\star(t,x)| \leq M_R^\star$, projection of $f(t,x)$ onto the interval $[-M_R^\star, M_R^\star]$ moves the logit closer to the pointwise minimizer and therefore cannot increase the conditional cross-entropy loss. Hence
$$L_{\mathrm{CE},2R}(Tf) \leq L_{\mathrm{CE},2R}(f).$$
Combining this with the contraction of the spatial-gradient seminorm yields
$$J_{\lambda,2R}(Tf) \leq J_{\lambda,2R}(f).$$

Applying this to the unique minimizer $f_{\lambda,2R}$, we see that $Tf_{\lambda,2R}$ is also a minimizer. By uniqueness,
$$Tf_{\lambda,2R} = f_{\lambda,2R},$$
and therefore
$$\|f_{\lambda,2R}\|_{L^\infty(C_{2R})} \leq M_R^\star = \|f^\star\|_{L^\infty(C_{2R})} \leq M_R.$$
This proves the claim. □

**Lemma C.3** (Bias on $C_{2R}$)**.** *Let $f_{\lambda,2R}$ be the unique minimizer from Lemma C.2. Define*

$$R_0^\star(t,x) := \Delta_x f^\star(t,x) + \nabla_x f^\star(t,x)\cdot\nabla_x \log\rho_t(x),$$

*and*

$$\mathcal{R}_0(R) := \int_{t_0}^T \|R_0^\star(t,\cdot)\|^2_{L^2(B_{2R},\mu_t)}\,dt.$$

*Let the boundary remainder be*

$$\mathfrak{b}_R := 2M_R\int_{t_0}^T\int_{\partial B_{2R}} \rho_t(x)|\partial_n f^\star(t,x)|\,dS(x)\,dt.$$

*Then*

$$\|f_{\lambda,2R}-f^\star\|^2_{L^2(C_{2R})} \le Cc_R^{-2}\mathcal{R}_0(R)\lambda^2 + Cc_R^{-1}\lambda\mathfrak{b}_R, \tag{78}$$

*and*

$$\|\nabla_x(f_{\lambda,2R}-f^\star)\|^2_{L^2(C_{2R})} \le Cc_R^{-1}\mathcal{R}_0(R)\lambda + C\mathfrak{b}_R. \tag{79}$$

*In particular, with*

$$\beta(R) := Cc_R^{-2}\mathcal{R}_0(R), \tag{80}$$

*we have*

$$\|f_{\lambda,2R}-f^\star\|^2_{L^2(C_{2R})} \le \beta(R)\lambda^2 + Cc_R^{-1}\lambda\mathfrak{b}_R, \tag{81}$$

*and*

$$\|\nabla_x(f_{\lambda,2R}-f^\star)\|^2_{L^2(C_{2R})} \le \beta(R)\lambda + C\mathfrak{b}_R. \tag{82}$$

*Moreover,*

$$\mathcal{R}_0(R) \le C(1+R)^K, \qquad \mathfrak{b}_R \le C(1+R)^K e^{-cR^2}, \tag{83}$$

*and therefore*

$$\beta(R) \le Ce^{CR}(1+R)^K. \tag{84}$$

*Proof.* Let

$$\delta_\lambda := f_{\lambda,2R} - f^\star.$$

Since $f_{\lambda,2R}$ minimizes $J_{\lambda,2R}$ over the convex set $\mathcal{H}_{M_R}(C_{2R})$, and since $f^\star \in \mathcal{H}_{M_R}(C_{2R})$, the variational inequality in the direction $f^\star - f_{\lambda,2R}$ gives

$$DJ_{\lambda,2R}(f_{\lambda,2R})[f^\star - f_{\lambda,2R}] \ge 0.$$

Equivalently,

$$\int_{t_0}^T\int_{B_{2R}} (\sigma(f_{\lambda,2R})-\eta_t)\delta_\lambda\,d\mu_t\,dt + 2\lambda\int_{t_0}^T\int_{B_{2R}} \nabla_x f_{\lambda,2R}\cdot\nabla_x\delta_\lambda\,d\mu_t\,dt \le 0. \tag{85}$$

The logistic curvature lower bound on $[-M_R, M_R]$ gives

$$\int_{t_0}^T\int_{B_{2R}} (\sigma(f_{\lambda,2R})-\eta_t)\delta_\lambda\,d\mu_t\,dt \ge c_R\|\delta_\lambda\|^2_{L^2(C_{2R})}. \tag{86}$$

Writing $f_{\lambda,2R} = f^\star + \delta_\lambda$, we obtain from (85) and (86)

$$c_R\|\delta_\lambda\|^2_{L^2(C_{2R})} + 2\lambda\|\nabla_x\delta_\lambda\|^2_{L^2(C_{2R})} \le -2\lambda\int_{t_0}^T\int_{B_{2R}} \nabla_x f^\star\cdot\nabla_x\delta_\lambda\,d\mu_t\,dt. \tag{87}$$

For each fixed $t$, Green's identity on $B_{2R}$ yields

$$-\int_{B_{2R}} \nabla_x f^\star\cdot\nabla_x\delta_\lambda\,\rho_t\,dx = \int_{B_{2R}} R_0^\star\delta_\lambda\,d\mu_t - \int_{\partial B_{2R}} \rho_t\delta_\lambda\partial_n f^\star\,dS. \tag{88}$$

Since both $f_{\lambda,2R}$ and $f^\star$ are bounded by $M_R$ on $C_{2R}$, we have

$$|\delta_\lambda| \le |f_{\lambda,2R}| + |f^\star| \le 2M_R.$$

Therefore, the boundary contribution in (88) is bounded by $\mathfrak{b}_R$, and Cauchy–Schwarz gives

$$c_R \|\delta_\lambda\|^2_{L^2(C_{2R})} + 2\lambda \|\nabla_x \delta_\lambda\|^2_{L^2(C_{2R})} \le 2\lambda \sqrt{\mathcal{R}_0(R)} \|\delta_\lambda\|_{L^2(C_{2R})} + 2\lambda \mathfrak{b}_R. \tag{89}$$

Applying Young's inequality,

$$2\lambda \sqrt{\mathcal{R}_0(R)} \|\delta_\lambda\|_{L^2(C_{2R})} \le \frac{c_R}{2} \|\delta_\lambda\|^2_{L^2(C_{2R})} + \frac{C\lambda^2}{c_R} \mathcal{R}_0(R),$$

we obtain

$$\frac{c_R}{2} \|\delta_\lambda\|^2_{L^2(C_{2R})} + 2\lambda \|\nabla_x \delta_\lambda\|^2_{L^2(C_{2R})} \le \frac{C\lambda^2}{c_R} \mathcal{R}_0(R) + 2\lambda \mathfrak{b}_R.$$

Dropping the gradient term gives

$$\|\delta_\lambda\|^2_{L^2(C_{2R})} \le C c_R^{-2} \mathcal{R}_0(R) \lambda^2 + C c_R^{-1} \lambda \mathfrak{b}_R,$$

which proves (78). Dropping instead the $L^2$-term and dividing by $2\lambda$ gives

$$\|\nabla_x \delta_\lambda\|^2_{L^2(C_{2R})} \le C c_R^{-1} \mathcal{R}_0(R) \lambda + C \mathfrak{b}_R,$$

which proves (79). The beta-form bounds (81)–(82) follow immediately from the definition of $\beta(R)$.

It remains to prove the growth estimates in (83). First, we control the mixture score. Since

$$p_t = (\alpha_t)_\# p_0 * \mathcal{N}(0, \sigma_t^2 I), \qquad q_t = (\alpha_t)_\# q_0 * \mathcal{N}(0, \sigma_t^2 I),$$

the same Gaussian-factorization argument used in Lemma C.1 gives

$$\|\nabla_x \log p_t(x)\| + \|\nabla_x \log q_t(x)\| \le C(1 + \|x\|),$$

uniformly over $t \in [t_0, T]$. Since

$$\nabla_x \log \rho_t(x) = \frac{p_t(x)}{p_t(x) + q_t(x)} \nabla_x \log p_t(x) + \frac{q_t(x)}{p_t(x) + q_t(x)} \nabla_x \log q_t(x),$$

we have

$$\|\nabla_x \log \rho_t(x)\| \le C(1 + \|x\|).$$

Together with Lemma C.1, which gives polynomial growth of the derivatives of $f^\star$ and a uniform bound on $\nabla_x f^\star$, this implies

$$|R_0^\star(t, x)| \le C(1 + \|x\|)^K.$$

Therefore

$$\mathcal{R}_0(R) = \int_{t_0}^T \|R_0^\star(t, \cdot)\|^2_{L^2(B_{2R}, \mu_t)}\, dt \le C(1 + R)^K.$$

Next, we bound the boundary remainder. By Lemma C.1,

$$|\partial_n f^\star(t, x)| \le \|\nabla_x f^\star(t, x)\| \le C.$$

Moreover, the compact-support VP representation implies the pointwise density bound

$$\rho_t(x) \le C \exp\{-c(\|x\| - R_\nu)_+^2\},$$

uniformly over $t \in [t_0, T]$. Hence, on $\partial B_{2R}$,

$$\rho_t(x) |\partial_n f^\star(t, x)| \le C e^{-cR^2}.$$

Multiplying by the surface area of $\partial B_{2R}$, the length of the time interval, and $M_R = C_M(1+R)$, we obtain

$$\mathfrak{b}_R \le C(1+R)^K e^{-cR^2}.$$

This proves (83). Finally, (84) follows from

$$\beta(R) = C c_R^{-2} \mathcal{R}_0(R), \qquad \mathcal{R}_0(R) \le C(1+R)^K,$$

and the growth bound $c_R^{-2} \le e^{CR}$, which follows from $M_R = C_M(1+R)$ and $c_R = (4\cosh^2(M_R/2))^{-1}$. $\square$

**Lemma C.4** (Approximation by the $R$-dependent clipped class)**.** *Under Assumption 5.1, for every $R \ge 1$ there exists a deterministic comparator*

$$f_{0,R} \in \mathcal{F}_{2R}$$

*such that*

$$\|f_{0,R} - f^\star\|^2_{H^{1,x}(C_{2R})} \le C(1+R)^K N^{-\frac{2(s-1)}{d+1}}. \tag{90}$$

*The comparator $f_{0,R}$ is fixed before observing the data.*

*Proof.* Recall the rescaled target

$$f_R^\star(t,\bar{x}) = f^\star(t, 2R\bar{x}), \qquad (t,\bar{x}) \in C_1.$$

By Lemma C.1 and the choice of $C_M$, we have

$$\|f_R^\star\|_{L^\infty(C_1)} \le M_R, \qquad \|\nabla_{\bar{x}} f_R^\star\|_{L^\infty(C_1)} \le M_R.$$

After enlarging $C_M$, if necessary, the stability constant from the spline quasi-interpolation construction in Appendix A is also absorbed into the envelope $M_R$. Therefore, applying Proposition A.4 on the fixed compact time-space domain $C_1 \subset \mathbb{R}^{d+1}$, there exists

$$f_R^{\mathrm{app}} \in \mathcal{F}_R$$

such that

$$\|f_R^{\mathrm{app}} - f_R^\star\|^2_{H^1_{t,\bar{x}}(C_1;\mathrm{Leb})} \le C_{\mathrm{app}} \|f_R^\star\|^2_{H^s_{t,\bar{x}}(C_1;\mathrm{Leb})} N^{-\frac{2(s-1)}{d+1}}. \tag{91}$$

Define the pulled-back comparator

$$f_{0,R} := \mathsf{P}_R f_R^{\mathrm{app}} \in \mathcal{F}_{2R}.$$

Let

$$e_R := f_R^{\mathrm{app}} - f_R^\star \quad \text{on } C_1, \qquad e := f_{0,R} - f^\star \quad \text{on } C_{2R}.$$

For $x \in B_{2R}$,

$$e(t,x) = e_R\left(t, \frac{x}{2R}\right), \qquad \nabla_x e(t,x) = \frac{1}{2R} \nabla_{\bar{x}} e_R\left(t, \frac{x}{2R}\right).$$

By the Gaussian lower-noise bound in Assumption 5.1, there exists $\rho_\star < \infty$, independent of $R$, such that

$$\sup_{t \in [t_0,T]} \sup_{x \in \mathbb{R}^d} \rho_t(x) \le \rho_\star.$$

Thus

$$\begin{aligned}
\|e\|^2_{H^{1,x}(C_{2R})} &= \int_{t_0}^T \int_{B_{2R}} \big(|e(t,x)|^2 + \|\nabla_x e(t,x)\|^2\big) \rho_t(x)\, dx\, dt \\
&\le \rho_\star \int_{t_0}^T \int_{B_{2R}} \big(|e(t,x)|^2 + \|\nabla_x e(t,x)\|^2\big)\, dx\, dt.
\end{aligned}$$

Changing variables $x = 2R\bar{x}$ gives

$$\int_{B_{2R}} |e(t,x)|^2\, dx = (2R)^d \int_{B_1} |e_R(t,\bar{x})|^2\, d\bar{x},$$

and

$$\int_{B_{2R}} \|\nabla_x e(t,x)\|^2\,dx = (2R)^{d-2}\int_{B_1} \|\nabla_{\bar{x}} e_R(t,\bar{x})\|^2\,d\bar{x}.$$

Since $R \ge 1$, $(2R)^{d-2} \le (2R)^d$. Hence

$$\|e\|^2_{H^{1,x}(C_{2R})} \le \rho_\star (2R)^d \|e_R\|^2_{H^{1,\bar{x}}(C_1;\mathrm{Leb})} \le \rho_\star (2R)^d \|e_R\|^2_{H^1_{t,\bar{x}}(C_1;\mathrm{Leb})}.$$

Combining this display with (91) gives

$$\|f_{0,R} - f^\star\|^2_{H^{1,x}(C_{2R})} \le C(2R)^d \|f_R^\star\|^2_{H^s_{t,\bar{x}}(C_1;\mathrm{Leb})} N^{-\frac{2(s-1)}{d+1}} = C\mathcal{A}_s(R) N^{-\frac{2(s-1)}{d+1}}.$$

It remains to control the $H^s$-norm of the rescaled target. For every $a + |\gamma| \le s$,

$$\partial_t^a D_{\bar{x}}^\gamma f_R^\star(t,\bar{x}) = (2R)^{|\gamma|} \partial_t^a D_x^\gamma f^\star(t, 2R\bar{x}).$$

By Lemma C.1,

$$\left|\partial_t^a D_x^\gamma f^\star(t,2R\bar{x})\right| \le C(1+R)^K \qquad (t,\bar{x}) \in C_1.$$

Since $C_1$ has finite Lebesgue volume, we obtain

$$\mathcal{A}_s(R) := (2R)^d \|f_R^\star\|^2_{H^s_{t,\bar{x}}(C_1;\mathrm{Leb})} \le C(1+R)^K.$$

□

**Lemma C.5** (Entropy of the pulled-back vector class)**.** *Let $\mathcal{F}_R = \mathcal{F}^{(d+1)}_{M_R}(L,W,S,B)$ be the $R$-dependent clipped-gradient class on $C_1 = [t_0,T] \times B_1$, with input dimension $d+1$, depth $L = O(1)$, sparsity $S = O(N)$, width $W$, and weight bound $B$. Define*

$$\mathcal{V}_{2R} := \{(h, \nabla_x h) : h = \mathsf{P}_R f,\ f \in \mathcal{F}_R\},$$

*with norm*

$$\|(u,v)\|_\infty := \|u\|_{L^\infty(C_{2R})} + \|v\|_{L^\infty(C_{2R})}.$$

*Then there exists a constant $C_{\mathrm{ent}} < \infty$, independent of $R, N, n, \lambda$, such that for every $0 < \varepsilon < 1$,*

$$\log \mathcal{N}\big(\varepsilon, \mathcal{V}_{2R}, \|\cdot\|_\infty\big) \le C_{\mathrm{ent}} N \log\left(\frac{BWM_R}{\varepsilon}\right). \tag{92}$$

*In particular, for the final choice $R = A\sqrt{\log n}$, the extra $\log M_R$ factor is absorbed into $\log(BWn)$.*

*Proof.* For $h = \mathsf{P}_R f$, we have

$$h(t,x) = f\left(t, \frac{x}{2R}\right), \qquad \nabla_x h(t,x) = \frac{1}{2R}\nabla_{\bar{x}} f\left(t, \frac{x}{2R}\right).$$

Since $R \ge 1$, an $\varepsilon$-cover of

$$\{(f, \nabla_{\bar{x}} f) : f \in \mathcal{F}_R\}$$

on $C_1$ induces an $\varepsilon$-cover of $\mathcal{V}_{2R}$ on $C_{2R}$. Thus it suffices to control the entropy of the derivative-augmented clipped network class on $C_1$.

The clipped-gradient class $\mathcal{F}_R$ is a subset of the clipped sparse ReLU$^3$ architecture with $O(S) = O(N)$ active parameters. On each fixed sparsity pattern, the network output and its first derivatives depend Lipschitzly on the nonzero parameters on the compact domain $C_1$. The Lipschitz constant is polynomial in $B, W$ and in the deterministic envelope $M_R$, while $L = O(1)$. The number of sparsity patterns contributes an additional $O(S\log W)$ term. A standard parameter-discretization argument therefore gives

$$\log \mathcal{N}\big(\varepsilon, \{(f,\nabla_{\bar{x}} f) : f \in \mathcal{F}_R\}, \|\cdot\|_\infty\big) \le C_{\mathrm{ent}} N \log\left(\frac{BWM_R}{\varepsilon}\right).$$

This proves (92). □

1210 **Lemma C.6** (Comparator-centered anisotropic oracle inequality on $C_{2R}$)**.** *Let $h_0 \in \mathcal{F}_{2R}$ be any deterministic comparator.*
1211 *Let*

$$\widehat{h}_R \in \arg\min_{h \in \mathcal{F}_{2R}} \widehat{J}_{\lambda,\mathcal{D}_{\mathrm{ext}},2R}(h)$$

1212 *be an empirical minimizer of the truncated empirical objective over $\mathcal{F}_{2R}$. Then, for every $t > 0$, with probability at least*
1213 *$1 - e^{-t}$,*

$$J_{\lambda,2R}(\widehat{h}_R) - J_{\lambda,2R}(f_{\lambda,2R}) \le C \left[ J_{\lambda,2R}(h_0) - J_{\lambda,2R}(f_{\lambda,2R}) + \Gamma_R \frac{N \log(BWM_R n) + t}{n} \right], \tag{93}$$

1214 *where*

$$\Gamma_R := c_R^{-1} + M_R + \lambda \left( \frac{M_R}{2R} \right)^2. \tag{94}$$

1215 *In particular, since $M_R = C_M(1+R)$ and $\lambda \le 1$,*

$$\Gamma_R \le Ce^{CR}(1+R)^K. \tag{95}$$

1216 *Proof.* For $h \in \mathcal{F}_{2R}$, define the population excess

$$E_R(h) := J_{\lambda,2R}(h) - J_{\lambda,2R}(f_{\lambda,2R}) \ge 0.$$

1217 For $h \in \mathcal{F}_{2R}$, define the comparator-centered excess loss

$$\psi_h(t,x,y) := \ell_{\mathrm{CE}}(y, h(t,x)) - \ell_{\mathrm{CE}}(y, h_0(t,x)) + \lambda \left( \|\nabla_x h(t,x)\|^2 - \|\nabla_x h_0(t,x)\|^2 \right).$$

1218 Then

$$\mathbb{E}[\psi_h] = J_{\lambda,2R}(h) - J_{\lambda,2R}(h_0) = E_R(h) - E_R(h_0),$$

1219 up to the fixed normalization of the time integral, which is absorbed into constants.

1220 Let

$$\bar{M}_R := \frac{M_R}{2R}.$$

1221 For every $h = \mathsf{P}_R f \in \mathcal{F}_{2R}$,

$$\|h\|_{L^\infty(C_{2R})} \le M_R, \qquad \|\nabla_x h\|_{L^\infty(C_{2R})} \le \bar{M}_R.$$

1222 Thus the same bounds hold for $h_0$.

1223 We first apply the Talagrand contraction lemma to the vector variable

$$(h, \nabla_x h).$$

1224 Indeed, if $h_1, h_2 \in \mathcal{F}_{2R}$, then, using the 1-Lipschitzness of the logistic loss and the spatial-gradient envelope

$$\sup_{h \in \mathcal{F}_{2R}} \|\nabla_x h\|_{L^\infty(C_{2R})} \le G_R, \qquad G_R := \frac{M_R}{2R},$$

1225 we have, pointwise in $(t,x,y)$,

$$\begin{aligned} |\psi_{h_1}(t,x,y) - \psi_{h_2}(t,x,y)| &\le |h_1(t,x) - h_2(t,x)| \\ &\quad + \lambda \left| \|\nabla_x h_1(t,x)\|_2^2 - \|\nabla_x h_2(t,x)\|_2^2 \right| \\ &\le |h_1(t,x) - h_2(t,x)| + 2\lambda G_R \|\nabla_x h_1(t,x) - \nabla_x h_2(t,x)\|_2. \end{aligned}$$

1226 Therefore, by the vector-valued Talagrand contraction lemma,

$$\begin{aligned} &\mathfrak{R}_n\big(\{\psi_h : E_R(h) + E_R(h_0) \le r\}\big) \\ &\quad \le C\,\mathfrak{R}_n\left(\{h - h_0 : E_R(h) + E_R(h_0) \le r\}\right) \\ &\qquad + C\lambda G_R\,\mathfrak{R}_n\left(\{\nabla_x h - \nabla_x h_0 : E_R(h) + E_R(h_0) \le r\}\right). \end{aligned} \tag{C.xx}$$

Here no time derivative is involved: the contraction is with respect to the spatial vector $(h, \nabla_x h)$, because the loss depends only on $h$ and $\nabla_x h$, not on $\partial_t h$.

Next, the localization $E_R(h) + E_R(h_0) \le r$ implies a localized anisotropic radius. By strong convexity of $J_{\lambda,2R}$,

$$E_R(h) \ge \frac{c_R}{2}\|h - f_{\lambda,2R}\|^2_{L^2(C_{2R})} + \lambda\|\nabla_x(h - f_{\lambda,2R})\|^2_{L^2(C_{2R})},$$

and the same bound holds for $h_0$. Hence

$$\|h - h_0\|^2_{L^2(C_{2R})} \le C c_R^{-1} r,$$

and

$$\lambda\|\nabla_x(h - h_0)\|^2_{L^2(C_{2R})} \le Cr.$$

Consequently,

$$\|h - h_0\|^2_{L^2(C_{2R})} + \|\lambda G_R(\nabla_x h - \nabla_x h_0)\|^2_{L^2(C_{2R})} \le C(c_R^{-1} + 1)r,$$

where we used $G_R \lesssim 1$ and $0 < \lambda < 1$.

Thus the contraction step reduces the problem to bounding the Rademacher complexity of a localized subset of the pulled-back vector class

$$\mathcal{V}_{2R} = \{(h, \nabla_x h) : h = \mathsf{P}_R f,\ f \in \mathcal{F}_R\},$$

with $L^2$-radius of order

$$\sqrt{(c_R^{-1} + 1)r}.$$

Applying Dudley's entropy integral and the entropy bound in Lemma C.5, we get

$$\begin{aligned}&\mathfrak{R}_n\big(\{\psi_h : E_R(h) + E_R(h_0) \le r\}\big)\\ &\quad\le \frac{C}{n} + \frac{C}{\sqrt{n}}\int_{1/n}^{C\sqrt{(c_R^{-1}+1)r}} \sqrt{N\log\left(\frac{BWM_Rn}{\varepsilon}\right)}\,d\varepsilon.\end{aligned} \tag{C.xx}$$

Evaluating the integral yields

$$\mathfrak{R}_n\big(\{\psi_h : E_R(h) + E_R(h_0) \le r\}\big) \le C\sqrt{\frac{(c_R^{-1} + 1)r\,N\log(BWM_Rn)}{n}},$$

after absorbing the lower-order $1/n$ term. This is the desired localized Rademacher bound. The envelope of the comparator-centered loss satisfies

$$|\psi_h| \le 2(\log(1 + e^{M_R}) + M_R) + 2\lambda\bar{M}_R^2 \le C\big(M_R + \lambda\bar{M}_R^2\big).$$

The variance bound is

$$\mathbb{E}[\psi_h^2] \le C\big(c_R^{-1} + \lambda\bar{M}_R^2\big)\big(E_R(h) + E_R(h_0)\big).$$

Let

$$U_R(h) := E_R(h) + E_R(h_0),$$

and define the normalized centered process

$$\widetilde{\psi}_h := \frac{\mathbb{E}[\psi_h] - \psi_h}{U_R(h) + r}.$$

The preceding envelope bound implies

$$\|\widetilde{\psi}_h\|_\infty \le \frac{2C(M_R + \lambda\bar{M}_R^2)}{r}.$$

Moreover, using

$$\mathrm{Var}(\psi_h) \le \mathbb{E}[\psi_h^2],$$

and the variance bound above, we obtain

$$\mathbb{E}[\widetilde{\psi}_h^2] \le \frac{C(c_R^{-1} + \lambda\bar{M}_R^2)U_R(h)}{(U_R(h) + r)^2} \le \frac{C(c_R^{-1} + \lambda\bar{M}_R^2)}{r}.$$

Thus the normalized class satisfies the boundedness and variance conditions required by Talagrand's concentration inequality. The peeling lemma applied to the local Rademacher bound gives

$$\mathfrak{R}_n\left(\left\{\frac{\psi_h}{U_R(h)+r} : h \in \mathcal{F}_{2R}\right\}\right) \le \frac{4\phi_R(r)}{r},$$

where

$$\phi_R(r) = C\sqrt{\frac{(c_R^{-1}+1)r\,N\log(BWM_Rn)}{n}}.$$

By symmetrization,

$$\mathbb{E}\sup_h \frac{\mathbb{E}[\psi_h]-\mathbb{E}_D[\psi_h]}{U_R(h)+r} \le \frac{8\phi_R(r)}{r}.$$

Therefore the same normalized-process and Talagrand concentration argument as in Theorem 3.5, applied with the localization functional $E_R(h)+E_R(h_0)$, yields the oracle inequality

$$E_R(\widehat{h}_R) \le C\left[E_R(h_0) + \Gamma_R\frac{N\log(BWM_Rn)+t}{n}\right],$$

where

$$\Gamma_R = c_R^{-1} + M_R + \lambda\bar{M}_R^2.$$

This proves (93). Finally, $M_R = C_M(1+R)$, $\bar{M}_R = M_R/(2R) = O(1)$, and $c_R^{-1} \lesssim e^{CR}$, so

$$\Gamma_R \le Ce^{CR}(1+R)^K.$$

$\square$

### C.3. Proof of Theorem 5.2

Let

$$\alpha := \frac{s-1}{(d+1)+2s-2}.$$

We decompose

$$\mathcal{E}(\widetilde{f}^{(R)}) := \int_{t_0}^T \|\widetilde{f}_t^{(R)} - f_t^\star\|_{H^1(\mu_t)}^2\,dt = \mathrm{Main}(R) + \mathrm{Tail}(R), \tag{96}$$

where

$$\mathrm{Main}(R) := \int_{t_0}^T \int_{B_R} \mathcal{L}(\widetilde{f}^{(R)}, f^\star)\,d\mu_t\,dt,$$

and

$$\mathrm{Tail}(R) := \int_{t_0}^T \int_{B_R^c} \mathcal{L}(\widetilde{f}^{(R)}, f^\star)\,d\mu_t\,dt.$$

**Step 1: Reduction of the main error to $C_{2R}$.** On $B_R$, $\chi_R \equiv 1$, and $\Pi_{2R}(x) = x$. Hence

$$\widetilde{f}^{(R)}(t,x) = \widehat{f}_R\Big(t, \frac{x}{2R}\Big) =: \widehat{f}_R^{\mathrm{pb}}(t,x), \qquad x \in B_R.$$

Therefore,

$$\mathrm{Main}(R) \le \int_{t_0}^T \int_{B_{2R}} \mathcal{L}(\widehat{f}_R^{\mathrm{pb}}, f^\star)\,d\mu_t\,dt. \tag{97}$$

**Step 2: The truncated ERM event.** Define

$$E_R := \left\{ \max_{1 \le i \le n} \|X_i\| \le 2R \right\}.$$

By Lemma C.1, the marginal distributions $p_t$ and $q_t$ have uniformly sub-Gaussian tails. Therefore there exist constants $C, c > 0$, independent of $n$ and $R$, such that

$$\sup_{t \in [t_0, T]} \max\{p_t(\{x : \|x\| > u\}), q_t(\{x : \|x\| > u\})\} \le C \exp\{-c(u - R_\nu)_+^2\}.$$

Conditioning on the sampled times and labels and applying a union bound,

$$\mathbb{P}(E_R^c \mid t_{1:n}, Y_{1:n}) \le \sum_{i=1}^{n} \mathbb{P}(\|X_i\| > 2R \mid t_i, Y_i) \le nC \exp\{-c(2R - R_\nu)_+^2\}.$$

For $R \ge R_\nu$, $(2R - R_\nu)_+ \ge R$. Hence

$$\mathbb{P}(E_R^c) \le nCe^{-cR^2}. \tag{98}$$

On $E_R$, $\Pi_{2R}(X_i) = X_i$, and for $h = \mathsf{P}_R f$,

$$\frac{\lambda}{4R^2} \|\nabla_{\bar{x}} f(t_i, X_i/(2R))\|^2 = \lambda \|\nabla_x h(t_i, X_i)\|^2.$$

Thus, on $E_R$, minimizing the projected empirical objective $\widehat{J}^{\text{proj}}_{\lambda, \mathcal{D}_{\text{ext}}, 2R}(f)$ over $f \in \mathcal{F}_R$ is equivalent to minimizing the truncated empirical objective over $h \in \mathcal{F}_{2R}$:

$$\widehat{J}_{\lambda, \mathcal{D}_{\text{ext}}, 2R}(h) := \frac{1}{n} \sum_{i=1}^{n} \left[ \ell_{\text{CE}}(Y_i, h(t_i, X_i)) + \lambda \|\nabla_x h(t_i, X_i)\|^2 \right] \mathbf{1}\{X_i \in B_{2R}\}. \tag{99}$$

Consequently, on $E_R$, $\widehat{f}^{\text{pb}}_R$ is an empirical minimizer of (99) over $\mathcal{F}_{2R}$.

**Step 3: Comparator-centered oracle inequality on $C_{2R}$.** Let $f_{\lambda, 2R}$ be the population minimizer over $\mathcal{H}_{M_R}(C_{2R})$, and recall

$$E_R(f) := J_{\lambda, 2R}(f) - J_{\lambda, 2R}(f_{\lambda, 2R}), \qquad f \in \mathcal{F}_{2R}.$$

By Lemma C.4, there exists a deterministic comparator

$$f_{0,R} \in \mathcal{F}_{2R}$$

such that

$$\|f_{0,R} - f^\star\|^2_{H^{1,x}(C_{2R})} \le C \mathcal{A}_s(R) N^{-\frac{2(s-1)}{d+1}} \le C(1+R)^K N^{-\frac{2(s-1)}{d+1}}. \tag{100}$$

Applying Lemma C.6 with $h_0 = f_{0,R}$ and $t > 0$, we obtain an event $\mathcal{E}_{\text{loc}}(R, t)$, with

$$\mathbb{P}(\mathcal{E}_{\text{loc}}(R, t)^c) \le e^{-t},$$

on which

$$E_R(\widehat{f}^{\text{pb}}_R) \le C \left[ E_R(f_{0,R}) + \Gamma_R \frac{N \log(BW M_R n) + t}{n} \right]. \tag{101}$$

Here

$$\Gamma_R = c_R^{-1} + M_R + \lambda \left( \frac{M_R}{2R} \right)^2,$$

and by Lemma C.6,

$$\Gamma_R \le Ce^{CR}(1+R)^K. \tag{102}$$

**Step 4: Bound the comparator excess.** We bound

$$J_{\lambda,2R}(f_{0,R}) - J_{\lambda,2R}(f_{\lambda,2R})$$

by comparing both terms to $f^\star$:

$$\begin{aligned} J_{\lambda,2R}(f_{0,R}) - J_{\lambda,2R}(f_{\lambda,2R}) \le J_{\lambda,2R}(f_{0,R}) - J_{\lambda,2R}(f^\star) \\ + J_{\lambda,2R}(f^\star) - J_{\lambda,2R}(f_{\lambda,2R}). \end{aligned}$$

First, since $f^\star$ is the pointwise Bayes logit, it minimizes the population cross-entropy risk. The logistic loss is $1/4$-smooth in the logit; hence

$$L_{\mathrm{CE},2R}(f_{0,R}) - L_{\mathrm{CE},2R}(f^\star) \le C\|f_{0,R} - f^\star\|^2_{L^2(C_{2R})}.$$

For the Sobolev penalty, using the pulled-back gradient envelope

$$\|\nabla_x f_{0,R}\|_{L^\infty(C_{2R})} \le \frac{M_R}{2R} \lesssim 1$$

and the uniform spatial-gradient bound for $f^\star$ from Lemma C.1, we get

$$\begin{aligned} &\lambda \left| \|\nabla_x f_{0,R}\|^2_{L^2(C_{2R})} - \|\nabla_x f^\star\|^2_{L^2(C_{2R})} \right| \\ &\quad \le C\lambda \|\nabla_x (f_{0,R} - f^\star)\|_{L^2(C_{2R})} \le C\left[ \|f_{0,R} - f^\star\|^2_{H^{1,x}(C_{2R})} + \lambda^2 \right]. \end{aligned}$$

Therefore, by (100),

$$J_{\lambda,2R}(f_{0,R}) - J_{\lambda,2R}(f^\star) \le C\mathcal{A}_s(R) N^{-\frac{2(s-1)}{d+1}} + C\lambda^2. \tag{103}$$

Second, since $f^\star$ minimizes the cross-entropy risk,

$$L_{\mathrm{CE},2R}(f^\star) - L_{\mathrm{CE},2R}(f_{\lambda,2R}) \le 0.$$

Let

$$\delta_\lambda := f_{\lambda,2R} - f^\star.$$

Then

$$J_{\lambda,2R}(f^\star) - J_{\lambda,2R}(f_{\lambda,2R}) \le \lambda \left( \|\nabla_x f^\star\|^2_{L^2(C_{2R})} - \|\nabla_x f_{\lambda,2R}\|^2_{L^2(C_{2R})} \right).$$

As in the proof of Lemma C.3, Green's identity gives the bound

$$J_{\lambda,2R}(f^\star) - J_{\lambda,2R}(f_{\lambda,2R}) \le C\left[ \beta(R)\lambda^2 + c_R^{-1}\lambda\mathfrak{b}_R + \lambda\mathfrak{b}_R \right].$$

Combining this with (103) yields

$$\begin{aligned} &J_{\lambda,2R}(f_{0,R}) - J_{\lambda,2R}(f_{\lambda,2R}) \\ &\quad \le C\left[ \mathcal{A}_s(R) N^{-\frac{2(s-1)}{d+1}} + \lambda^2 + \beta(R)\lambda^2 + c_R^{-1}\lambda\mathfrak{b}_R + \lambda\mathfrak{b}_R \right]. \end{aligned} \tag{104}$$

**Step 5: Convert the oracle bound to $H^{1,x}$-error on $C_{2R}$.** By the strong convexity inequality on $C_{2R}$, for every $f \in \mathcal{F}_{2R}$,

$$E_R(f) \ge \frac{c_R}{2}\|f - f_{\lambda,2R}\|^2_{L^2(C_{2R})} + \lambda\|\nabla_x (f - f_{\lambda,2R})\|^2_{L^2(C_{2R})}.$$

Consequently,

$$\|f - f_{\lambda,2R}\|^2_{H^{1,x}(C_{2R})} \le C(c_R^{-1} \vee \lambda^{-1}) E_R(f).$$

For the final choice $R = A\sqrt{\log n}$ and $\lambda \asymp n^{-\alpha}$, we have

$$c_R^{-1} \lesssim e^{CR} = n^{o(1)},$$

so $\lambda^{-1}$ dominates $c_R^{-1}$ for all sufficiently large $n$. Thus, on $\mathcal{E}_{\mathrm{loc}}(R,t)\cap E_R$, using (101) and (104),

$$\|\widehat{f}_R^{\mathrm{pb}} - f_{\lambda,2R}\|^2_{H^{1,x}(C_{2R})} \le \frac{C}{\lambda}\left[\Gamma_R \frac{N\log(BWM_Rn)+t}{n} + \mathcal{A}_s(R)N^{-\frac{2(s-1)}{d+1}} + \lambda^2 + \beta(R)\lambda^2 + c_R^{-1}\lambda\mathfrak{b}_R + \lambda\mathfrak{b}_R\right].$$

Using Lemma C.3 to pass from $f_{\lambda,2R}$ to $f^\star$, we obtain

$$\int_{t_0}^T \int_{B_{2R}} \mathcal{L}(\widehat{f}_R^{\mathrm{pb}}, f^\star)\, d\mu_t\, dt \le C\left[\frac{\Gamma_R}{\lambda}\frac{N\log(BWM_Rn)+t}{n} + \frac{\mathcal{A}_s(R)}{\lambda}N^{-\frac{2(s-1)}{d+1}} + \lambda + \beta(R)\lambda + (c_R^{-1}+1)\mathfrak{b}_R\right]. \tag{105}$$

By (97), the same bound holds for $\mathrm{Main}(R)$.

**Step 6: Tail error.** On the annulus $B_{2R}\setminus B_R$,

$$\widetilde{f}^{(R)}(t,x) = \chi_R(x)\widehat{f}_R^{\mathrm{pb}}(t,x),$$

and therefore

$$\nabla_x\widetilde{f}^{(R)}(t,x) = (\nabla_x\chi_R)(x)\widehat{f}_R^{\mathrm{pb}}(t,x) + \chi_R(x)\nabla_x\widehat{f}_R^{\mathrm{pb}}(t,x).$$

Since

$$|\chi_R|\le 1, \qquad \|\nabla_x\chi_R\|_\infty \lesssim R^{-1},$$

and

$$\|\widehat{f}_R^{\mathrm{pb}}\|_{L^\infty(C_{2R})} \le M_R, \qquad \|\nabla_x\widehat{f}_R^{\mathrm{pb}}\|_{L^\infty(C_{2R})} \le \frac{M_R}{2R} \lesssim 1,$$

we have

$$|\widetilde{f}^{(R)}(t,x)| + \|\nabla_x\widetilde{f}^{(R)}(t,x)\| \le C(1+R), \qquad x\in B_{2R}\setminus B_R.$$

Together with the value and gradient bounds for $f^\star$ from Lemma C.1, this gives

$$\mathcal{L}(\widetilde{f}^{(R)}, f^\star)(t,x) \le C(1+R)^K, \qquad x\in B_{2R}\setminus B_R.$$

Therefore, using the tail bound in Lemma C.1, there exist constants $C, K, c_{\mathrm{ann}}>0$ such that

$$\int_{t_0}^T\int_{B_{2R}\setminus B_R} \mathcal{L}(\widetilde{f}^{(R)}, f^\star)\, d\mu_t\, dt \le C(1+R)^K e^{-c_{\mathrm{ann}}(R-R_\nu)_+^2}. \tag{106}$$

On $B_{2R}^c$, $\widetilde{f}^{(R)}=0$. Hence

$$\mathcal{L}(\widetilde{f}^{(R)}, f^\star) = |f^\star|^2 + \|\nabla_x f^\star\|^2.$$

By Lemma C.1, this is bounded by $C(1+\|x\|)^K$. The tail-moment bound in Lemma C.1 gives constants $C, K, c_{\mathrm{out}}>0$ such that

$$\int_{t_0}^T\int_{B_{2R}^c} \mathcal{L}(\widetilde{f}^{(R)}, f^\star)\, d\mu_t\, dt \le C(1+R)^K e^{-c_{\mathrm{out}}(2R-R_\nu)_+^2}. \tag{107}$$

Combining (106) and (107), and taking $R\ge 2R_\nu$, we have

$$(R-R_\nu)_+ \ge R/2, \qquad (2R-R_\nu)_+\ge R.$$

Thus, setting

$$c_{\mathrm{tail}} := \min\{c_{\mathrm{ann}}/4, c_{\mathrm{out}}\} > 0,$$

and enlarging $C, K$ if necessary, we obtain

$$\mathrm{Tail}(R) \le C(1+R)^K e^{-c_{\mathrm{tail}}R^2}. \tag{108}$$

**Step 7: Choosing $N, \lambda, R$.** Choose

$$N \asymp n^{\frac{d+1}{d+1+2s-2}}, \qquad \lambda \asymp n^{-\alpha}, \qquad \alpha = \frac{s-1}{d+1+2s-2}. \tag{109}$$

Then

$$\frac{N \log(BWM_R n)}{n} \lesssim \lambda^2 \log n, \qquad N^{-\frac{2(s-1)}{D}} \lesssim \lambda^2. \tag{110}$$

By Lemma C.1,

$$\mathcal{A}_s(R) \le C(1+R)^K,$$

and by Lemmas C.3 and C.6,

$$\beta(R) \le Ce^{CR}(1+R)^K, \qquad c_R^{-1} \le Ce^{CR}, \qquad \Gamma_R \le Ce^{CR}(1+R)^K.$$

Substituting these estimates into (105) and using (110), with $t = 2\log n$, gives

$$\mathrm{Main}(R) \le Ce^{CR}(1+R)^K (\log n) n^{-\alpha} + Ce^{CR}(1+R)^K \mathfrak{b}_R. \tag{111}$$

Since

$$\mathfrak{b}_R \le C(1+R)^K e^{-cR^2},$$

the boundary term is exponentially small for the choice of $R$ below. Combining (111) with (108), we obtain

$$\mathcal{E}(\widetilde{f}^{(R)}) \le Ce^{CR}(1+R)^K (\log n) n^{-\alpha} + Ce^{CR}(1+R)^K e^{-cR^2}. \tag{112}$$

Finally choose

$$R = A\sqrt{\log n},$$

with $A > 0$ sufficiently large so that $R \ge 2R_\nu$ for all sufficiently large $n$, the truncation-event failure probability in (98) is at most $n^{-2}$, and the exponential remainder in (112) is negligible relative to the main term. On the event

$$\mathcal{E}_{\mathrm{loc}}(R, 2\log n) \cap E_R,$$

whose probability is at least $1 - 2n^{-2}$, and hence at least $1 - 3n^{-2}$, we have

$$\mathcal{E}(\widetilde{f}^{(R)}) \le Ce^{C\sqrt{\log n}} (\log n)^K n^{-\alpha}. \tag{113}$$

Since

$$e^{C\sqrt{\log n}} (\log n)^K = n^{o(1)},$$

it follows that for every $\varepsilon > 0$, there exists $C_\varepsilon < \infty$ such that

$$\mathcal{E}(\widetilde{f}^{(R)}) \le C_\varepsilon n^{-\alpha+\varepsilon}. \tag{114}$$

This completes the proof. □

# D. Supplementary Experiment Results

## D.1. Synthetic Experiments

We provide details for synthetic source–target pairs here.

- **Rotated ridge.** In the $(t, s)$-coordinates the source and target are mixtures of two Gaussians with identical covariance but different offsets along the $t$-axis: the source has components centered at $\pm a$ and the target at $\pm b$ (with $b > a$), while the $s$-coordinate is shared. The pair $(t, s)$ is then rotated by an angle $\mathcal{P} = 30°$ into the observed $(x_1, x_2)$-space. This creates an anisotropic "ridge" structure where $p$ and $q$ differ primarily along a rotated onedimensional direction.

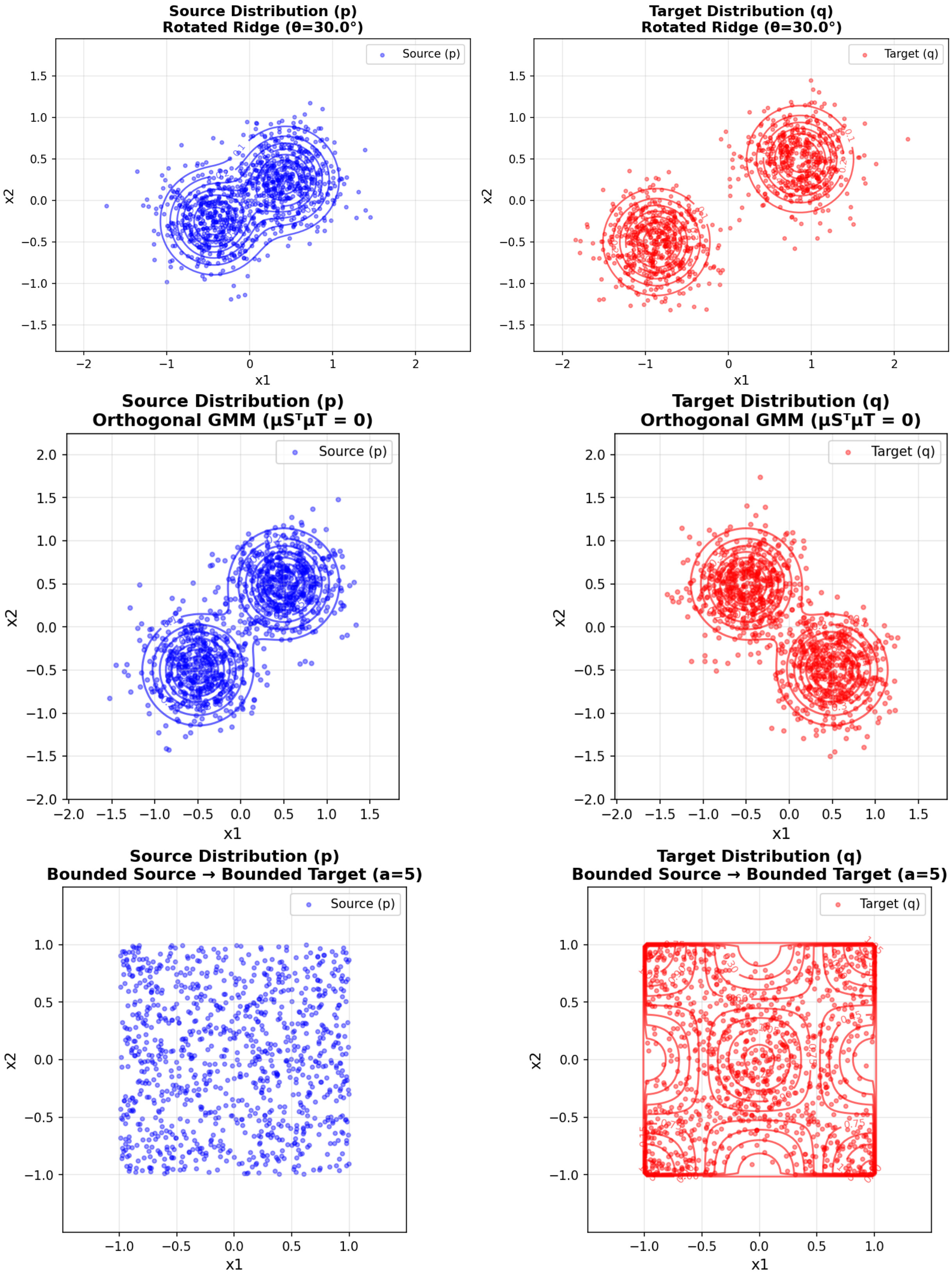


*Figure 4.* Three simulation distributions (left column: source $p$, right column: target $q$). **Top:** Same-line GMM with rotated ridge structure. **Middle:** Orthogonal GMM with misaligned source and target components. **Bottom:** Bounded source-to-target distribution with compact support.

- **Orthogonal GMM.** Both $p$ and $q$ are symmetric two-component Gaussian mixtures with spherical covariance. The source means are $\pm\mu_S$, the target means are $\pm\mu_T$, and we choose $\mu_S^\top \mu_T = 0$, i.e. the source and target mixture axes are orthogonal. This case is adopted in Ouyang et al. (2024) and probes how well the estimator adapts when the main directions of variation in the source and target are misaligned.

- **Bounded source to bounded target.** The source $p$ is the uniform distribution on the square $[-1, 1]^2$. The target $q$ is obtained by smoothly reweighting $p$ with a bounded, oscillatory density ratio $r(x) = \frac{q(x)}{p(x)} = c\big(1 + b\cos(\pi x_1)\cos(\pi x_2)\big)$, where the parameters are chosen so that $r(x) \in [1/a, a]$ with $a = 5$. This yields a non-Gaussian, compactly supported setting with a spatially varying but uniformly bounded density ratio.

Together, these three examples cover rotated lowdimensional shifts, orthogonal mixture structure, and bounded non-Gaussian modulation, and hence provide a diverse testbed for gradient-of-ratio estimation.

### D.2. WGF experiments

**Setting details.** When a large number of unlabeled target samples $x_q^{(k)}$ are available, (Courty et al., 2016) proposed learning an optimal transport map to align the joint source and target distributions ($P_{XZ}$ and $Q_{XZ}$), and subsequently training a classifier on the aligned source samples. Building on this framework, (Liu et al., 2023) utilized WGF for sample alignment. Their method evolves particles $x_t$ to minimize KL$[p_t, q]$ between the evolving particle-label distribution $p_t$ (initialized with source samples $\mathcal{D}_{p_t} := \{(x_t^{(i)}, y_p^{(i)})\}_{i=1}^{n_p}$) and the target density $q$. Upon convergence after $T$ iterations, a classifier is trained on the transported source samples $\{(x_T, y_q)\}$ to predict target labels.

Implementing WGF requires estimating the score difference. While (Liu et al., 2023) employed a local method known as "local linear interpolation" (LL) for this estimation, our proposed approach utilizes a classification-based method with Sobolev regularization to estimate the score difference globally.

### D.3. ECG experiments

**Details for two datasets.** PTB-XL contains 21,837 clinical 12-lead ECG recordings of 10-second duration from 18,885 patients, annotated with 71 diagnostic statements in a multi-label setting. The ICBEB2018 dataset consists of 6,877 12-lead ECG recordings with durations ranging from 6 to 60 seconds, each labeled into one of nine diagnostic classes, which form a subset of the PTB-XL label space. We randomly select 10% of the ICBEB2018 samples as the limited target dataset via stratified sampling to preserve label proportions. All ECG signals are resampled to a frequency of 100 Hz.

**Synthetic quality and training speed evaluation for diffusion models.** We evaluate the quality of the generated samples of the target task using the standard Fréchet Inception Distance (FID) metric (Heusel et al., 2017). The FID score measures the Wasserstein-2 distance between the distributions of real and synthetic data within the feature space of an xresnet1d50 classifier (Strodthoff et al., 2020) pre-trained on the target task.

The quantitative results are summarized in Table 6. Our proposed TGDP-SOB not only achieves the best FID of 8.097 but also significantly reduces model parameters to 2.8M and training time to 30 minutes, offering a superior balance between generation quality and computational efficiency.

*Table 6.* The effectiveness of Soblev Penalization on ECG benchmark under synthetic quality.

| Method | FID | Parameters | Training Time |
|---|---|---|---|
| Vanilla Diffusion | 11.171 | 50.2M | 1h |
| Finetune Generator | 8.415 | 50.2M | 40min |
| TGDP | 8.100 | **2.8M** | **30min** |
| **TGDP-SOB** | **8.097** | **2.8M** | **30min** |